Trenitalia
ETR1000 VZI-6

Conventional Train Control

Revision	Author	Date	Revision Description
00.1	R. Schiavo	09/07/2023	Draft Release Generated from DOORS module /Platform.-.Zefiro-Europe/.. /2F_09.03.-.Conventional Train Control-VF.-.Zefiro-Europe/Conventional Train Control-VF.CONCEPT, current 6.0 doc 3ECP413719-0903 reviewed for VZI-6
00	R. Schiavo	25/07/2023	First Emission
1.1	R. Schiavo	20/09/2023	Encoder with digital output has been replaced by an encoder with 4-20mA analog output
1.00	R.Schiavo	21/09/2023	Release Emission
2.1	R.Schiavo	04/04/2024	New Requirements .HR13-.HR52 (DNRA Alarm) Modified Requirement .1522
2.0	R.Schiavo	29/04/2024	Emission - BL1 baseline
3.1	R.Schiavo	08/08/2024	Modified Requirements: .HR17,.HR18,.HR19,.HR20,.HR53,.HR45,.HR46,.HR50 (DNRA)
3.2	R.Schiavo	29/08/2024	Modified requirement: .1522 - Deleted gear box / axle Overtemperature in C2 serious fault list TRS.155 Speed limit ATP cut out 80->90 km/h
3.3	R.Schiavo	04/09/2024	Modified requirement: .1522 – rejected previous change gear box / axle Overtemperature are C2 serious fault list
3.0	R. Schiavo	16/09/2024	Emission – BL2 baseline
4.1	G. De Marco	13/01/2025	Add section 3.1.26 (call for aid); add requirements HR53,HR54,HR55,HR56,HR57
4.2	R. Schiavo	14/01/2025	Added section “Train Speed Reduction / Train stopping due to Alarms” par 3.1.7 Modified req .HR24, .1220, .1572, .238,. 1572,.HR53,.HR54,.HR55 Modified 2F_09.03.Zefiro-Europe.TRS.155 come da CR_025 Added req. .HR 58, .HR59 (from FRS Bogie and FRS Brake),. HR60,.HR61,.HR62,.HR63,.HR64,.HR65
4.0	R. Schiavo	16/01/2025	Emission – BL3 baseline

5.1	P. Buda	30/05/2025	Modified classification serious faults par. 3.1.15.1, after Customer diagnostic system technical review, modifed par. 3.1.7.1. Modified the following req: .HR17, .HR18, .HR19, .HR20, .HR23, .HR53, .HR58, .1517, .1522, .1525, .1526, .1527, .1528, .1572, .1595 Corrected wrong numbering in par. 3.1.26 (Executive call for aid) for req. .HR53→ .HR66
5.2	G. Spina	03/06/2025	Multiple Unit requirements update: Modified requirements: 850, 336, 340, HR13, HR14, HR16, HR26, HR39, HR43, HR50, 155, 1519, 1580, 1238,1263 ,236,432,1613 -Added requirements: HR100, HR101,HR102, HR105, , HR106, HR108, HR110,HR121, Deleted req: 1390 Updated Function Design: 3.1.6.1, 3.1.7.1, 3.1.13.1
5.3			
5.0	**G. Spina**	**03/06/2025**	**Emission – BL4 baseline**

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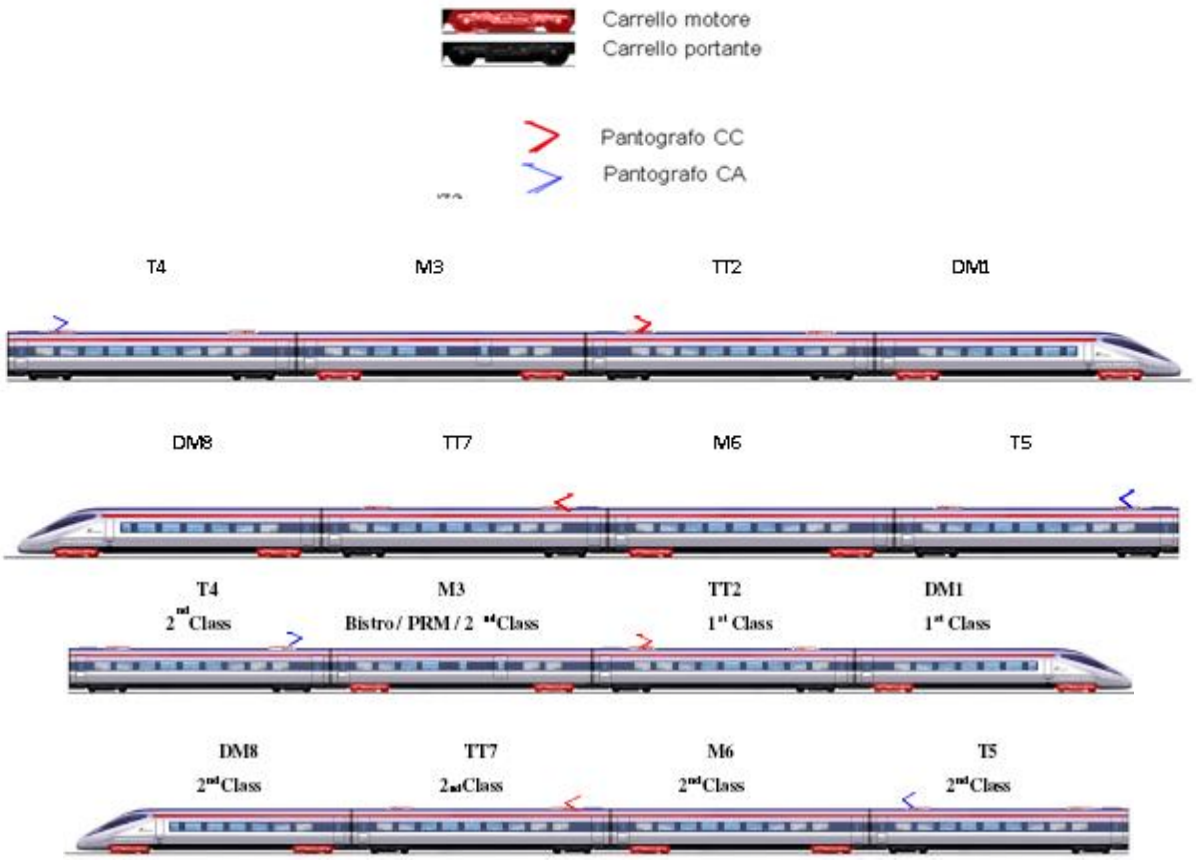
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1 Introduction

This document shall provide a description of the functionality for the ETR1000 VZI6 Train Control System including ordinary train control (OTC) by conventional relay logic and train lines, TCMS electronics and Sub-system control unit.

1.1 Train Configuration

Each ETR1000 train consists of eight cars in a fixed formation including a driver's cab at each end:



- DM = End driving motor car with driver's cab
- TT = Intermediate trailer car with transformer
- M = Intermediate driving motor car
- T = Intermediate trailer car

1.2 Definitions and abbreviations

ATP Automatic Train Protection
BCU Brake control unit
CDU Central and Diagnostic Control Unit
CCU Central Control UnitCD Circuit Diagram
DBV Discharge Brake Valve
DIS Drivers Indicated Speed
DNRA Detection of non rotating axle
DSD Drivers safety device
ETB Ethernet Train Backbone
EVENT Diagnostics event that will be stored in the diagnostics database on the CCU-D
HVAC Heating Ventilation and air condition

HV High voltage
HW Hardware
HW-HMI Hardware Human Machine Interface, e.g. buttons, switches and indication lamps.
I/O Input/output
LCB Line Circuit Breaker
LED Light Emitting Diode
MCM Motor converter module
MDCU Monitoring and Diagnostic Control Unit
MOCU/C/L/M/A Control unit Motor/Line/Chopper/Auxiliary converter
NVRAM Non Volatile Random Access Memory
OTC Ordinary train control(relay logic/hardware by circuit diagram)
PCU Propulsion Control Unit
RIO Remote Digital input/Output
SFRS System Functions Requirement Specification
SIL Safety Integrity Level
TBD To be defined
TCMS Train Control and Management System
TL Train lines
TOD Train Operator Display
TSI Train Specification for Interoperability
UIC Union Internationale des Chemin (International Union of Railways)
WSP Wheel slide protection

2 System Overview

2.1 Train Control Principles

The train will have a redundant two bus battery system that will affect all other electrical system/functionalities assembled.
The Train Control System is a combination of

A) **Ordinary Train Control** (OTC) comprising relay logic and train control lines
B) **TCMS electronics**, comprising hardware, software and communication networks
C) **Sub-system control unit**

The safety integrity level (SIL) is the major design constraint on where and how a function is realised, in the Train Control System, using a safe computer in the TCMS electronics and/or by the OTC relay logic.
All safety critical and safety related functions will be controlled by OTC or a safe computer.
The identified vital functions are realised using redundant train line pairs.
Train lines that if faulty will stop operation are possible to by-pass.
All vital functions related to TSI requirements shall be duplicated or protected to cope with a fire incident in a way that the train is always able to perform an evacuation drive.
The TCMS electronics will be used for diagnostics. It will also be used to control functions.

2.2 System Configuration

Train supervision and indications:

All safety critical and safety related functions will be controlled or supervised via OTC or a safe computer.
The identified vital functions are realised using redundant or fire safe train lines.
Train lines that if faulty will stop operation are possible to by-pass.

All vital functions will be duplicated to cope with a fire incident in a way that the train is always able to drive with limited performance (emergency operation).

The TCMS will be used for diagnostics and monitoring. It will also be used to control functions.

3 Functional requirements

3.1 Train Control

3.1.1 Cab Activation/Deactivation

The cab is activated by the driver or the crew personal moving the key switch on the Active position. This function works only if the train has been activated. Note: The train is activatedafter the pushbutton in the electrical cabinet in DM-car has been pushed and the Train Activation procedure did not fail.

A single Train Line is provided to inform that one cab is active (in single or multiple configuration)

3.1.1.1 *Design*

Cab Activation
Interlock must be implemented in order to prevent two cabs from being activated at the same time. If another cab activation attempt is done while a cab is activated, an advisory message is generated by TCMS.
In case two driver's key switch are rotated in active position when the train is deactivated, the TCMS cannot monitor the inconsistency since the 110 Vdc is not available. If in this condition the train active procedure is performed, none cabs must be activated and a fault need to be sent to the TOD and the "TCMS busy" lamp needs to be illuminated steadily.

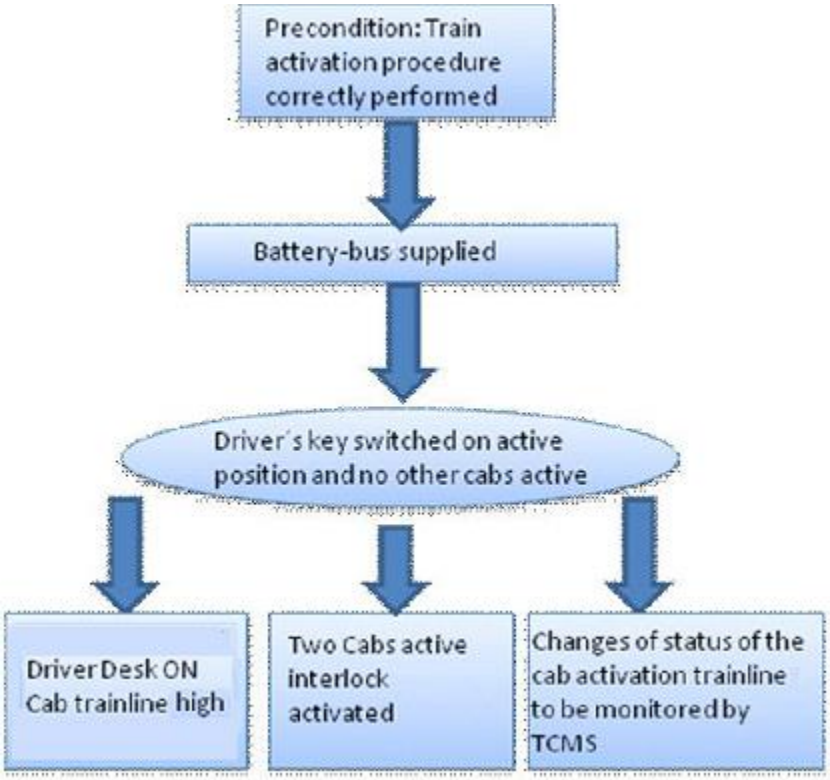
The TCMS supervises the status of the Cab Active Train line and of each desk key.
In case of any failure or inconsistency (eg two cabs active at the same time) traction block will be set until the train is back in normal condition or the block is reset using the TOD in maintenance mode

Cab Deactivation
The cab shall not be deactivated when speed >0km/h (note train is not standstill). If the key is turned when the train is running the Emergency Brake shall be applied and the line voltage shall be deactivated (LCB open and panto down)

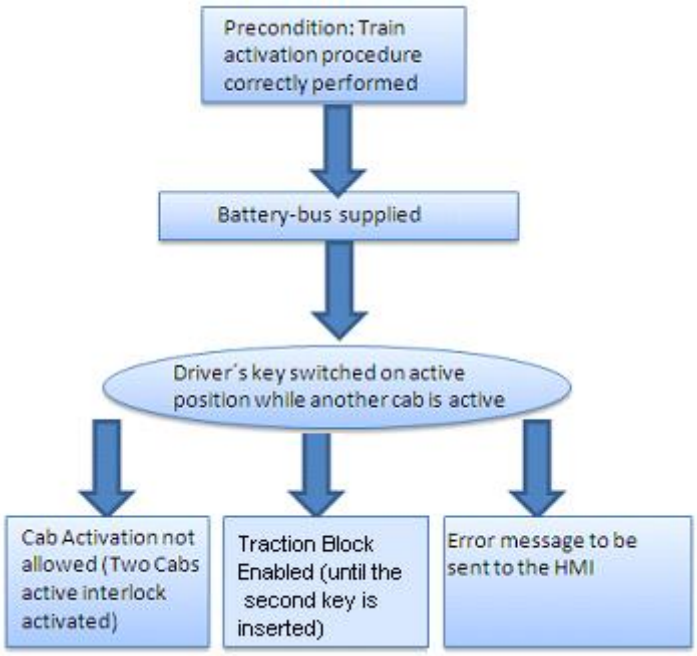
When the driver de-activates the cab then the proper shut down sequence for line voltage will be performed unless Parking mode is activated (see Parking mode section).

The status of the cab activation trainline is supervised by TCMS.

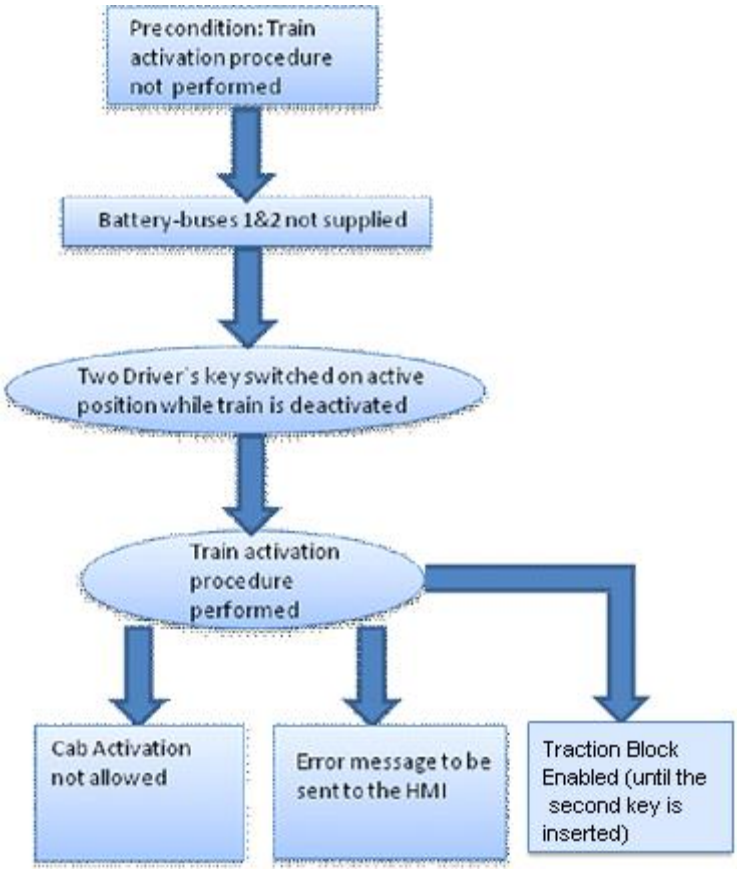
Cab Activation



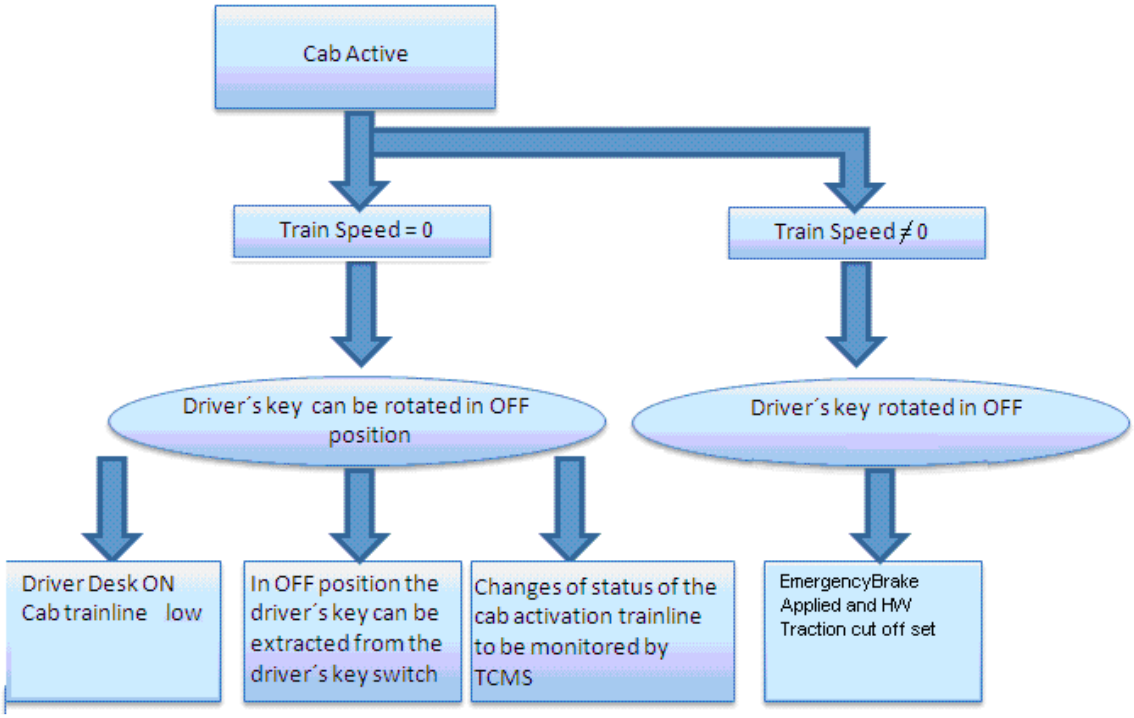
Cab Activation not allowed a)



Cab Activation not allowed b)



Cab Deactivation



3.1.1.1.1 Normal condition

A) Cab Activation: In normal condition driver rotate the driver´s key on the active position and the cab trainline shall be high.
B) Cab Deactivation: In normal condition driver rotate the driver´s key on the off position and the cab trainline shall be low.

3.1.1.1.2 Failure cases

1. the cab trainline is not high after the driver´s key has been turned on active position. Train will be immobilised.
2. the cab trainline is not low after the driver´s key has been turned on OFF position. Driving the train from opposite cab will not be possible.
3. Cab is activated when no driver´s key is in the active position.
4. Fault of one cab activation trainline

3.1.1.1.3 Failure detection

1. Performed by the driver (driver desk not active)
2. Performed by the driver (driver desk still active)
3. In both both cabs are active a fault message will be displayed on the TOD
4. Fault message on the TOD

3.1.1.1.4 Failure handling

1. Retire or don´t put the vehicle in service. Immediate corrective maintenance required.
2. Retire or don´t put the vehicle in service. Immediate corrective maintenance required.
3. Maintenance needed before putting the vehicle in service again.
4. Maintenance needed at the end of the service.

3.1.1.2 *Requirements*

ID	2F_09.03.Zefiro-Europe.TRS.347
Description	Two Position: 0 = Cab off (stable position) 1 = Cab activated (stable position). It is possible to extract the key only in OFF position
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1007	Cab shall be activated/de-activate by means of two signals "Cab Activated" and "Cab Off". The signals are managed by the driver by means of dedicated key switch (2 stable position switch).	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 1
2F_09.03.Zefiro-Europe.CONCEPT.1109	Cab is activated/de-activate by means of two signals "Cab Activated" and "Cab Off". The signals are managed by the driver by means of dedicated key switch (2 stable position switch).	Derived Requirement	4S_08.07.01.-.Drivers Desk		N/A
2F_09.03.Zefiro-Europe.CONCEPT.953	If the cab is attempted to be de-activated by the driver's key when train speed > 0km/h the Emergency Brake Loop shall be opened (EB applied) and the Traction Safe line shall be de-energized	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1279	If the cab is attempted to be de-activated by the driver's key when train speed > 0km/h an advisory event shall be displayed on the TOD.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

ID	2F_09.03.Zefiro-Europe.TRS.174
Description	Cab activation is made by means of the driver's key.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL

2F_09.03.Zefiro-Europe.CONCEPT.1006	Cab is activated when the signal "Cab Activated" is high (this is done by the driver moving the key on the Cab activated stable position).	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 1
2F_09.03.Zefiro-Europe.CONCEPT.1342	A cab shall be considered as activated when the 'Cab activated' input is set OR all the following conditions apply for at least 3 seconds: a) The "Any cab active' input is set b) No "Cab activated" input is set anywhere in the train c) The drivers key is in active position	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1596	In case a second cab is attempted to be enabled or disabled, TCMS shall consider first activated cab (key inserted and rotated) as really enabled This is valid also in case of multiple configuration	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2

ID	2F_09.03.Zefiro-Europe.TRS.175
Description	Activation of a second cab in a train shall be prevented.
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.948	Interlock shall be implemented in order to prevent two cabs from being activated at the same time.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.1436	If another cab is attempted to be activated (desk key turned) then Traction Block shall be set by TCMS.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.950	Any cab activation attempt in another non active cab shall be indicated in the cab where the activation is attempted.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A
2F_09.03.Zefiro-Europe.CONCEPT.951	The status of the 'driver's key' input, 'cab active' input (LCA and LCA-R relays) and the 'any cab activated trainline' input shall be supervised by TCMS for plausibility. If an inconsistency is detected an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1253	The 'Cab active' position of the driver's key shall be indicated to TCMS. The status of Cab Active relays (LCA) shall be provided to TCMS reading the series of NC and NO contacts separately. The status of the "any cab actived" trainlines shall be indicated to TCMS.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

ID	2F_09.03.Zefiro-Europe.TRS.222
Description	Sufficient brake shall be applied e.g. by opening the emergency brake loop in case the cab is deactivated. This shall avoid roll back in case of not occupied cab.
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.206	When no cab is active in the train the emergency brake loops shall be opened (Emergency Brake Applied).	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		**SIL 2**
2F_09.03.Zefiro-Europe.CONCEPT.1254	If there is no active cab in the train and the Emergency brake is not ordered (EB loop is energised) then an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1418	If there is no active cab in the train then Traction block shall be applied.	Derived Requirement	4S_09.03.05.-.TCMS Software		**SIL 0**

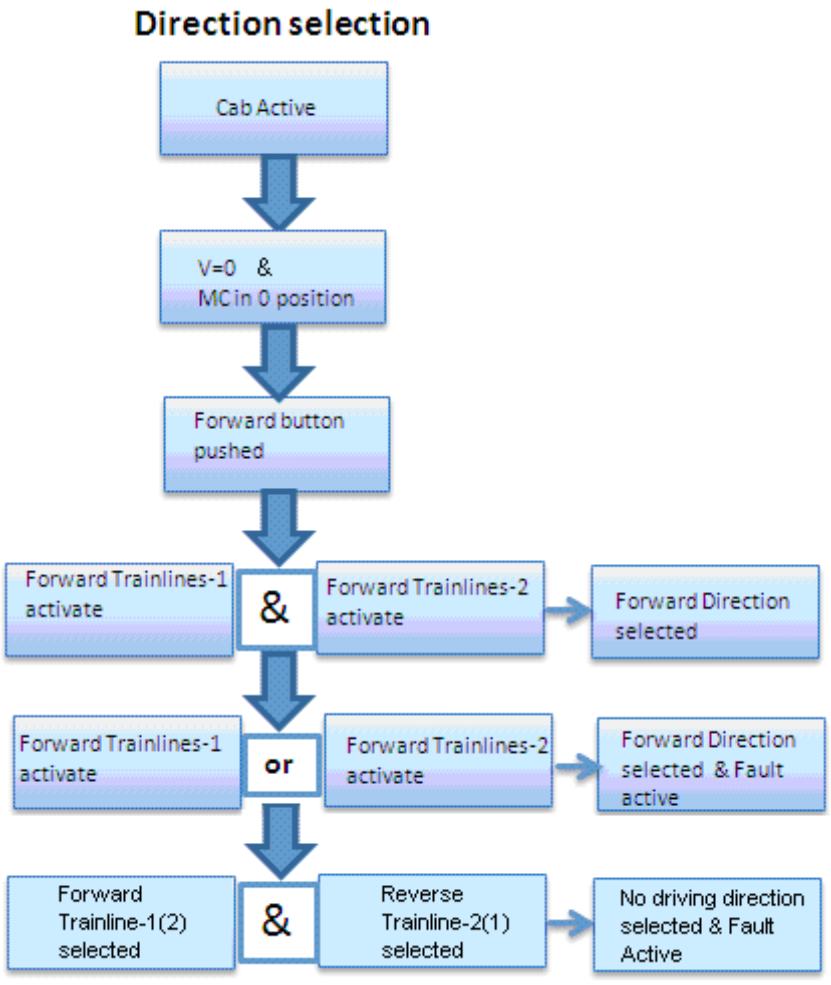
3.1.2 Demand of Driving Direction

3.1.2.1 *Design*

The driving direction signal is in normal operation transferred by trainlines. At failure a redundant parallel trainline is used.

It will be not possible to pass directly from Forward to Reverse direction. The driver must go from forward to neutral by pushing the neutral push-button at train standstill and with the master controller in position 0. Then from Neutral the **Reverse** direction can be selected by pushing the reverse push-button, similar behaviour need to be foreseen when it is needed to go from Reverse to Forward.

The TCMS will be used for monitoring. The forward/reverse trainlines will be wired into the MOCUM for motor converter modules to control the driving direction.



Same is valid for reverse direction selection

When driver select Reverse Direction speed limitation is applied.

3.1.2.1.1 Normal condition

a) After pushing the Forward pushbutton the required direction is selected
b) After pushing the Reverse pushbutton the required direction is selected

3.1.2.1.2 Failure cases

a) After pushing the Forward pushbutton the required direction is not selected
b) After pushing the Reverse pushbutton the required direction is not selected
c) After pushing the Forward(Reverse) pushbutton the opposite direction is selected

3.1.2.1.3 Failure detection

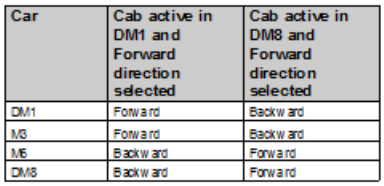
–
– Performed by the driver (Forward pushbutton is not lit after selection)
–
  – Performed by the driver (Reverse pushbutton is not lit after selection)
–
  – Performed by the driver

3.1.2.1.4 Failure handling

– Retire or don´t put the vehicle in service
– Retire or don´t put the vehicle in service
– Retire or don´t put the vehicle in service

3.1.2.2 *Requirements*

ID	2F_09.03.Zefiro-Europe.TRS.230
Description	Driving in reverse direction by system failure shall be avoided.
Safety level	SIL 1

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.965	It shall not be possible to change direction directly from forward to reverse or from reverse to forward. Neutral must be select before selecting direction by pushing the neutral push-button at train standstill and when the master controller is not in traction position. The direction can then be selected.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 1
2F_09.03.Zefiro-Europe.CONCEPT.437	If change of driving direction is attempted when the train is not standstill or when master controller is in traction position, an advisory event shall be generated.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.850	The driving direction shall be sent to propulsion control via trainlines in the whole train(also in multiple unit) according to the schema below:  If Neutral is selected on the driver's desk neither forward or backward shall be sent to the Propulsion control.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 1

Car	Cab active in DM1 and Forward direction selected	Cab active in DM8 and Forward direction selected
DM1	Forward	Backward
M3	Forward	Backward
M6	Backward	Forward
DM8	Backward	Forward

Car	Cab active in DM1 and Reverse direction selected	Cab active in DM8 and Reverse direction selected
DM1	Backward	Forward
M3	Backward	Forward
M6	Forward	Backward
DM8	Forward	Backward

2F_09.03.Zefiro-Europe.CONCEPT.848	Propulsion control shall supervise the direction signals received by trainlines interfaced with MOCU/M and take actions according to below table: Forward Trainline 1 / Forward Trainline 2 / Reverse Trainline 1 / Reverse Trainline 2 / Forward / Reverse / No driving direction / Fault indication 0 0 0 0 x 0 0 0 1 x x 0 0 1 0 x x 0 0 1 1 x 0 1 0 0 x x 0 1 0 1 x x 0 1 1 0 x x 0 1 1 1 x x 1 0 0 0 x x 1 0 0 1 x x 1 0 1 0 x x 1 0 1 1 x x 1 1 0 0 x 1 1 0 1 x x 1 1 1 0 x x 1 1 1 1 x x	Derived Requirement	4S_05.-.Propulsion		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1112	An event shall be displayed on the TOD if reverse direction has been selected	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

ID	2F_09.03.Zefiro-Europe.TRS.148
Description	Change of direction shall only be allowed when: 1) at standstill (v=0) 2) master controller in position "0"
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.515	The "forward direction" pushbutton, "reverse direction" pushbutton, "neutral" pushbutton shall be connected to TCMS.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1440	After the cab activation it shall not be possible to select any direction if the Master Controller is in Traction position. If a direction is selected while the Master controller is in Traction the TCMS shall apply Traction Block until the handle is moved in "0" or "braking" position.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1280	The status of below input shall be sent to TCMS from each PCU in the train: 1) Forward trainline input 1 2) Forward trainline input 2 3) Reverse trainline input 1 4) Reverse trainline input 2	Derived Requirement	4S_05.-.Propulsion		N/A

ID	2F_09.03.Zefiro-Europe.TRS.149
Description	Maximum reverse speed shall be limited to 40km/h.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.69	If driving in reverse direction has been demanded, the train speed shall be limited to 40km/h, the value shall be adjustable as parameter stored in the CCU.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_09.03.Zefiro-Europe.TRS.61
Description	When a cab will be activated this cab will be defined as head of the train. From this results the normal driving direction (forward direction). For shunting trips it is be possible to select reverse direction.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1009	The head of the train is defined according to the cab activated.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

ID	2F_09.03.Zefiro-Europe.TRS.62
Description	The active cab shall always define the forward direction. The driving drection shall be actively chosen by the driver using a "forward" or a "Reverse" button in the driver's desk.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.972	Forward direction of the train shall always be determined according to the active cab.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

ID	2F_09.03.Zefiro-Europe.TRS.351
Description	Direction selector pushbuttons Forward:Lit when selected, Neutral: Lit when selected (neutral selected by default when at the desk activation) Backward: Lit when selected
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.969	Three buttons are available on the direction selection panel: Forward: Lit when selected Neutral: Lit when selected (neutral selected by default when at the desk activation) Backward: Lit when selected	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.03.Zefiro-Europe.CONCEPT.971	When the cab is activated "neutral" shall be set	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1344	When the cab is activated an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1398	When the cab is activated and direction is not "neutral" a Fault message shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1415	If no valid direction is set Traction block shall be applied.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1416	If the direction is changed at speed >0 km/h then Traction block shall be applied. The Traction block shall be reset when master control handle is not in traction position AND (the evaluated direction is equal to real direction OR train speed = 0.)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.1.3 Traction/Brake reference

3.1.3.1 *Design*

3.1.3.1.1 Master Controller and Speed Lever

Each cab is equipped with a Master Controller and a Speed Lever controller.

Master Controller: Two redundant encoders with 4-20mA analog output are used in the Master controller to determine the demanded traction/brake reference.
The encoders are connected directly to TCMS

A set of microswitch is included in the controller to provide interfaces with:
1) Traction/Not Brake trainlines (see Emergency Operation)
2) Emergency Brake Loop
3) Emergency operation mode

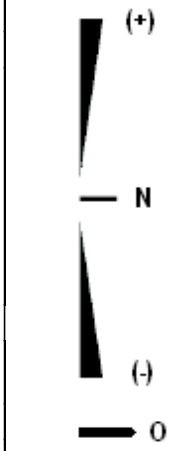
(*)	#Pos	Position [%]	Description
	6	T 100 %	*Maximum Traction Effort*
	5	T Min %	*Minimum Traction Effort*
	4	N	***Traction** can only be applied by **pressing the knob** on top of handle*
	3	B Min %	*Minimun Braking Effort*
	2	B 100 %	*Maximum Braking Effort*
	1	EB	*Emergency Braking Position*

Figure 1 Master Controller Lever positions

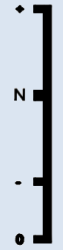
LEVER POSITION	POSITION IN DEGREES	ANALOG OUTPUT SIGNAL (mA)	TOLLERANCE (mA)
Rapid Brake	-50°	4,00	+/- 0,5 mA
Max Braking	-42°	5,39	
Min Braking	-8°	11,30	
Coasting	0°	12,70	
Min Traction	8°	14,09	
Max Traction	42°	20,00	

Figure 2 analog output signal -> Master Controller lever position

The Master controller also provides a signal to the vigilant device pushing the top of the handle
Speed Lever: The speed lever can be used by the driver to select a speed reference. The train will then be limited to this speed by TCMS.
The speed lever positions are connected directly to TCMS

(*)	#	Position	Description
	4	(+)	*Spring-loaded return to Position 3*
	3	N	*Predefined position:* *To **increase** target speed push forward the lever and press the knob on the top of the handle* *To **decrease** target speed, **no specific action** required*
	2	(-)	*Spring-loaded return to Position 3.*
	1	0	*Predefined position, semi-locked. ASC off*

Switching table for Speed controller

Micro-Switch ? / Position ?		S101	S102	S103	S104	S105	S106
	Speed +	0	0	0	0	1	1
	N	0	0	1	1	1	1
	Speed -	0	0	1	1	0	0
	0	1	1	1	1	1	1

3.1.3.1.2 Normal operation

In normal operation either the Master controller or the Speed Lever can be used to adjust the amount of requested traction/braking demand relevant to available traction braking effort.

The driver can change the mode to a speed set reference to drive the train to a specific speed. The Master controller will also be used to request the maximum amount of power that will be used to achieve the set speed reference.

During normal operation the driver can change from traction to speed mode by means of the dedicated speed lever acknowledgment button which shall be pressed while the master controller is in coasting position ("N") and the speed lever in “N” position (other combination will not allow the change from traction mode to speed mode).
When the speed mode has been activated, the actual speed of the train shall be used as reference speed. The driver can adjust the reference speed in step of 1 km/h or in step of 5 km/h. It depends on the time the driver keep the speed lever in + or – position.
In case of multiple unit composition TCMS send the traction/braking demand to remote TCMS via ETBN network.

3.1.3.1.3 Failure cases (master controller)

1) Traction is demanded when not requested
2) Braking is demanded when not requested
3) Traction is not demanded when requested
4) Braking is not demanded when requested
5) One encoder is not working
6) Both encoders fault

3.1.3.1.4 Failure detection (master controller)

1) By the driver
2) By the driver
3) By the driver
4) By the driver
5) Message on the TOD
6) Not admissible difference between the signals currents values (>1mA). Message on the TOD

3.1.3.1.5 Failure handling (master controller)

1. Operate the 'Master controller back up mode' switch (sealed) and use the speed lever to demand a traction reference. Switch Brake mode to "Indirect brake" and use DBV to set brake reference. Maintenance needed at the end of the service.
2. Operate the 'Master controller back up mode' switch (sealed) and use the speed lever to demand a traction reference. Switch Brake mode to "Indirect brake" and use DBV to set brake reference. Maintenance needed at the end of the service.
3. Operate the 'Master controller back up mode' switch (sealed) and use the speed lever to demand a traction reference. Switch Brake mode to "Indirect brake" and use DBV to set brake reference. Maintenance needed at the end of the service.
4. Operate the 'Master controller back up mode' switch (sealed) and use the speed lever to demand a traction reference. Switch Brake mode to "Indirect brake" and use DBV to set brake reference. Maintenance needed at the end of the service.
5. Complete the service. Maintenance needed at the end of the service
6. Operate the 'Master controller back up mode' switch (sealed) and use the speed lever to demand a traction reference. Switch Brake mode to "Indirect brake" and use DBV to set brake reference. Maintenance needed at the end of the service

3.1.3.1.6 Failure cases (Speed lever)

1) Speed of train is greater than the speed set reference
2) Speed of train is lower than the speed set reference

3.1.3.1.7 Failure detection (Speed lever)

1) By driver or ATP demands brake if the train speed is over the allowed track speed
2) By the driver

3.1.3.1.8 Failure handling (Speed lever)

1) The train will try to regulate the speed by braking to the speed set reference. The driver can also select a brake position of the Master controller. If the speed set reference is the same or close to that of the ATP speed limit the ATP will brake the train. Maintenance required.
2) The driver can increase the power requested by the Master controller or remove the speed set reference mode and drive only with the Master controller

The following table reports the master Controller handling in Normal and degraded contions:

Traction TL	NotBrake TL	Encoder 1	Encoder 2	Delta>1mA	Diagnostic	Running mode	Rate Reference	Degraded Condition
1	1	Tra	Tra		-	Tra	Min(enc1,Enc2)	
0	0	Brk	Brk		-	Brk	Max(enc1,Enc2)	
0	1	Coa	Coa		-	Coa	0	
1	1	Tra	Tra	x	Enc1, Enc2, MC in fault	TT	0	MC back up mode
0	0	Brk	Brk	x	Enc1, Enc2, MC in fault	Brk	Max(enc1,Enc2)	MC back up mode
0	1	Coa	Coa	x	Enc1, Enc2, MC in fault	TT	0	MC back up mode
-	-	Brk (Tra/Coa)	Tra/Coa (Brk)	x	Enc1, Enc2, MC in fault	Brk	Enc1 (Enc2)	MC back up mode
-	-	Brk	Brk	x	Enc1, Enc2, MC in fault	Brk	Max(enc1,Enc2)	MC back up mode
-	-	Tra/Coa	Tra/Coa	x	Enc1, Enc2, MC in fault	TT	0	MC back up mode
1	1	Coa	Tra		Enc1 in fault	Tra	Enc2	
0	1	Tra	Coa		Enc1 in fault	Coa	0	
0	0	Coa/Tra	Brk		Enc1 in fault	Brk	Enc2	
1	1	Tra	Coa		Enc2 in fault	Tra	Enc1	
0	1	Coa	Tra		Enc2 in fault	Coa	0	
0	0	Brk	Coa/Tra		Enc2 in fault	Brk	Enc1	
1	1	Coa	Coa		Enc1, Enc2, MC in fault	TT	0	MC back up mode
1	1	Brk	Brk		Enc1, Enc2, MC in fault	Brk	Max(enc1,Enc2)	MC back up mode
.	0	Tra/Coa	Tra/Coa		Enc1, Enc2, MC in fault	TT	0	MC back up mode
0	1	Tra	Tra		Enc1, Enc2, MC in fault	TT	0	MC back up mode

Table 1 Master Controller Handling

3.1.3.2 *Requirements*

ID	2F_09.03.Zefiro-Europe.TRS.197
Description	Two controllers, the ordinary master controller and the speed controller work independently.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1010	Master controller and Speed Set Lever are two independent units.	Derived Requirement	4S_08.07.01.-.Drivers Desk		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1196	Master controller and Speed Set Lever are two independent units.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

ID	2F_09.03.Zefiro-Europe.TRS.336
Description	The traction demand level shall be distributed from the active cab to all PCUs along the train.
Safety level	SIL 1

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1149	The positive effort shall be equally shared among the available traction inverter while the negative effort shall be distribute according to the Blending law (Blending law description is not part of this SFRS).	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.HR100	In multiple unit configuration the traction/braking demand level set in enabled cab shall be sent to remote TCMS via ETBN network. The TCMS of remote train shall forward the requested demand to PCU	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.794	The traction demand shall be forwarded and distributed to the PCU in the motor cars.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.795	If traction demand is > 0% it shall be indicated to the PCU that train is in traction.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

2F_09.03.Zefiro-Europe.CONCEPT.796	If traction demand is < 0% it shall be indicated to the PCU that train is in braking.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.797	If friction brake is applied in the train then the PCU shall be notified.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_09.03.Zefiro-Europe.TRS.338
Description	No traction force shall be applied if not demanded.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.792	If the Master controller is not in a traction position or the EmergencyOperation Mode trainline is not active or traction is not demanded by the Speed set lever in Master controller back-up mode or the speed mode is not enabled then the traction demand to all PCU's in the train shall be set to 0%.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_09.03.Zefiro-Europe.TRS.359
Description	The master controllers brake position has priority even if the speed controller is not in “0” position. This because braking has always priority.
Safety level	N/A

The train can be driven in Power reference mode or Speed set reference mode.
Power reference mode is when only the power reference from the master control controller demands a specific amount of traction or brake percentage based on available traction or braking effort.
Speed set reference mode is when the speed is selected by the speed set lever. The power for achieving the speed set reference is then demanded by the power reference position of the Master controller.

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1155	It shall not be possible to select a power or speed set reference mode if traction block is active and the master controller is not in the coast position.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1156	In Power reference mode operation both requested and applied torques shall be displayed on TOD. In Speed reference mode the requested speed (target speed) shall be displayed on TOD	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1157	The control of the traction/brake demand shall be automatically set to Power reference mode at cab activation.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1158	To select speed set reference mode the speed lever shall be moved to "N" position while the master controller is in the 'N' position (coasting).	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1159	When 'speed reference mode' is selected the current train speed shall be the used as the target one	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1532	After the speed control mode has been enabled, the traction position of the master controller shall limit the power reference to achieve the required speed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1160	To return to power reference mode from speed set reference mode, the speed lever shall be moved in "0" position.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1161	Switching between modes can be done at any time, regardless of speed.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1408	When speed reference mode is reset Traction block shall be applied until master controller is in coast or brake position.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1162	The power or speed set reference shall be shown as a flashing “0” on the TOD when there is a	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A

	traction block but the requested torque is not negative.				
2F_09.03.Zefiro-Europe.CONCEPT.1163	If a neutral section or separation section is detected the power brake reference shall be set to 0 or a negative value.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1164	When neutral section or separation section is reset the power brake reference shall be resumed (traction is allowed).	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1165	The TOD shall show the selected speed set reference or power reference during the whole neutral section or separation section passage even though the power brake reference is temporary set to 0 or a negative value.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1166	The master controllers brake position has priority even if the speed controller is not in “0” position.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_09.03.Zefiro-Europe.TRS.187
Description	A speed controller for speed set (cruise control) shall be provided.Positions 4: increase speed set point 3: neutral 2: decrease set poing 1: Off (speed set disabled) The speed set point (target speed) shall be integrated over time. The time base depends on the time the lever is in the + or - position. I.e in the + position >1s=fast increase, <1s = slow increase in the - position >1s=slow decrease, >1s = fast decrease
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1167	The speed lever positions and the acknowledment pushbutton shall be evaluted by TCMS.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1395	The positions of the speed lever and the acknowledment pushbutton shall be connected to TCMS.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1168	It shall never be possible to select a speed reference that is greater than any speed limitation that is active	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1345	With the vehicle in speed mode and speed limitation applied, if the driver tries to increase the speed regardless the speed limit by means of the speed lever a warning message shall be displayed on the TOD.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1169	Speed set reference is increased by three different increase times of the speed lever. 1 - 1.If speed lever is set in '+' position → Speed set reference shall increase 1km/h per trigger. 2 - If speed lever is set in '+' position for >500ms and <3s → Speed set reference shall increase 5km/h every 500ms. 3 - If speed lever is set in '+' position for >5s → Speed set reference shall increase 5km/h fast, 0 to 360km/h in about 9 sec.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1170	Speed set reference is decreased by three different decrease times of the speed lever. • 1.If speed lever is set in '-' position → Speed set reference shall decrease 1km/h per trigger. • If speed lever is set in '-' position for >500ms and <3s → Speed set reference shall decrease 5km/h every 500ms. • If speed lever is set in '-' position for >5s → Speed set reference shall decrease 5km/h fast, 0 to 360km/h in about 9 sec.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

2F_09.03.Zefiro-Europe.CONCEPT.1171	The speed set reference shall be shown on the TOD display.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1175	If speed set reference mode is activated the TCMS automatically regulates the speed to the requested speed reference.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1174	Speed set reference shall be automatically set to 0 if no traction is available and speed is v = 0 km/h.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_09.03.Zefiro-Europe.TRS.188
Description	A master controller for traction and braking effort demand shall be provided. The driver sets the demand of traction or braking power in different levels of controller positions.
Safety level	SIL 1

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1148	There shall be 2 encoders in the master controller with 4-20mA outputs	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1176	The 4-20mA outputs of each encoder shall be connected to TCMS	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1177	The traction/brake power reference shall increase linearly over the full movement of the master controller brake handle.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1178	The shall be a plausibility check between the two encoders and the status of the traction/brake trainlines.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1194	If a encoder is considered faulty event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1180	Both the requested and applied traction/brake power shall be shown on TOD.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A

ID	2F_09.03.Zefiro-Europe.TRS.370
Description	It shall be possible to perform a test of the Master controller from the TOD. In the test each position of the Master controller shall be tested by manually moving the Master controller based on instructions provided on the TOD.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1032	Master controller test shall be activated from the TOD only when logged in as a maintainer	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1033	The TOD shall present a screen that instructs the driver to put the Master controller in each position. For each position the test logic shall verify that the Master controller indicates the required position.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1031	**Master controller test is possible according to the following conditions:** Master controller test shall only be possible at stand still v=0. Master controller test shall set 100% service brake as long as the test is running.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1034	The Master controller test shall be possible to restart any time during the test sequence by pressing a button on the TOD.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1035	The two encoders are tested during the Master controller Test If one of the two encoders is faulty the faulty encoder shall be deselected by the software and not used. If both encoders are faulty traction block shall be applied as long as the cab with the faulty Master controller is activated.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.HR04	The Master controller test consists of checking the encoder analog output in pre-defined positions. If the measured analog output signal	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

	differs more than ±0.5 mA from the forecasted value then the encoder is considered faulty				
2F_09.03.Zefiro-Europe.CONCEPT.1036	**Master Controller handle test results:** Test done, both encoders are OK Encoder A faulty Encoder B faulty Master controller out of order	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1393	The result of the master Controller Test shall be displayed on the TOD	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1394	An event shall be set detailing the result of the Master Controller test	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

ID	2F_09.03.Zefiro-Europe.TRS.361
Description	The master controller functionality shall be redundant.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1023	The Master controller is equipped with two redundant encoders and a set of trainlines to acquire the traction/braking effort information in both normal and emergency operation	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

In normal condition both encoders work, TCMS acquires and use the information from both encoders in order to provide a reliable traction/braking effort request.

ID	2F_09.03.Zefiro-Europe.TRS.371
Description	In case of Master Controller fault, Master controller back-up mode (sealed latched switch) can be selected. In this mode the driver can drive the train by using the Speed controller.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1181	If at any time, one encoder output differs more than ±1.0mA from the other, for a longer time than 5 seconds, both encoderes are considered faulty and an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.HR05	if both encoders are faulty and both encoder positions are in the traction range a traction cut out is applied	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.HR06	if both encoders are faulty and both encoder positions are in the brake range a brake is applied with the maximum reference between the 2 encoders	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.HR07	if both encoders are faulty and both encoder positions are in the coasting range a traction cut out is applied	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.HR08	if both encoders are faulty and an encoder position is in coasting or traction range and the other is in brake range a brake is applied with the reference from the encoder in brake position	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.HR09	if Traction TL and Not Brake TL configuration is not coherent with only one encoder position that encoder is considered faulty	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.HR10	if an encoder is considered faulty an event is generated	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.HR11	if Traction TL and Not Brake TL configuration is not coherent with both encoders position the Master controller is considered faulty and a traction cut out is applied	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.HR12	if the Master controller is considered faulty an event is generated	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1183	It shall always be the highest brake position or the lowest traction request from either contact	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

	set of the Master controller that is used to determine Master controller position.				

If both encoders are faulty or there is a mechanical fault with the Master controller (Master controller Major failure) Traction block shall be applied till standstill and "back-up mode" enabled.

Urgency Brake will also applied till standstill in case of both encoders faulty according requirement 2F_05.02.Zefiro-Europe.CONCEPT.1055

When the train is standstill the following instruction shall be displayed:
- Operate 'Master controller backup mode' switch
- Switch to Indirect Brake mode

The train need to be repaired soon after the end of service.

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1188	If the "Master controller backup mode" switch has been operated AND Train Braking mode is set on "Indirect Brake" Then Master controller back up mode shall be considered ACTIVE	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1578	Traction block shall be applied if MCH backup mode is turned to backup position. Traction block shall be reset when 1. Train Braking mode is set on "Indirect Brake" OR 2. MCH backup mode switch in normal position OR 3. Emergency operation mode is active	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_09.03.Zefiro-Europe.CONCEPT.1541	If both encoders are faulty the Traction block shall be applied until the train is stopped and "back-up mode" is enabled: 1. Brake selector moved in "indirect brake" position 2. Back up mode switch moved in "enabled" position	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1533	"Master controller backup mode' switch can be operated only when the train is in standstill	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1185	If both encoders are faulty then an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1186	If the "Master controller backup mode' switch is operated then an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1536	An indication of the traction set point selected through speed lever shall be displayed on TOD.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1538	An indication that Master controller back up mode is enabled shall be displayed in TOD.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1189	If the 'Master controller backup mode' is active then the speed lever shall be used to select traction reference only.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1190	When the speed lever is in the '+' position in Master controller back up mode then the traction set point shall increase linearly (till 100%) as long as the lever is kept in this position.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1191	When the speed lever is in the '-' position in Master controller back up mode then the traction set point shall be reduced linearly (till 0%) as long as the lever is kept in this position.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1192	If the speed lever is in the 'N' position Master controller back up mode then last traction setpoint shall be used as reference.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1537	When the speed lever is in the '0' position Master controller back up mode traction reference is set directly to 0%	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_09.03.Zefiro-Europe.TRS.209
Description	The ultimate position of the traction/brake shall cause emergency braking.
Safety level	SIL 4

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1028	Last braking position of the Master Controller shall cause the application of the Emergency Brake.	Derived Requirement	4S_09.02.09.-.Conventional Train Control – Circuits		SIL 4
2F_09.03.Zefiro-Europe.CONCEPT.1195	An event shall be displayed on the TOD if there is an Emergency Brake application due to the Master Controller.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1282	If the Master controller is in Emergency brake position and Emerrgency brake is not applied within 2 seconds (Urgency brake loop is deenergised) then an event shall be generated.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

3.1.4 Traction Safe

3.1.4.1 *Function Design*

The traction safe is a function to ensure that the train is ready to move before enabling the traction.
Traction shall be enabled only when all conditions for traction are fulfilled.
By energizing the traction safe train line, the traction will be enable.

To prevent the traction, the traction safe trainline shall be deenergized.

If positive traction is to be inhibited traction block shall be set (TCMS via IP Communication).
If positive traction and electrodynamic brake are inhibited, the “Traction safe” trainline shall be de-energized.

When the traction safe trainline is de-energized, to recover the traction and the electrodynamic brake is necessary to energize the traction safe trainline and to put the master controller in Coasting before to command traction or electrodynamic brake

The traction safe can be in any case overridden via HW (All doors are not closed and locked from door, Low brake pipe pressure, Traction safe loop de-energised functions) and via SW (All doors are not closed and locked from doors, Low brake pipe pressure, External power supply is connected, External air supply is connected, Traction safe loop de-energised). Further description are included into the Traction Safe Override function described into this SFRS.

3.1.4.1.1 Normal condition

The traction safe trainline shall energized, and so traction shall be enabled only when all conditions for traction are fulfilled.

3.1.4.1.2 Failure cases

1. Traction is not released even if all the condition for releasing the function are satisfied.
2. Traction Safe trainline is energized even if same conditions are not satisfied

3.1.4.1.3 Failure detection

- Done by the driver (no traction provided with the master controller in traction position)
- Alarm to be set on the TOD

3.1.4.1.4 Failure handling

- If traction is not released, override function can be used to clear the track. Safety procedure need to be defined to drive the vehicle with override function active. Train need to be put out of the service as soon as it is possible.
- Stop the train as soon as possible. Towing/Restriction operation requested

3.1.4.2 *Requirements*

ID	2F_09.03.Zefiro-Europe.TRS.340
Description	Traction shall only be possible in the whole train (also in mltiple unit configuration) when all conditions for traction release are fulfilled. For that case there shall be a hard-wired train line "traction safe".
Safety level	SIL 2

ID	2F_09.03.Zefiro-Europe.TRS.373
Description	Traction shall be disabled if the pressure in the brake pipe is less then 3,5 bars.
Safety level	SIL 2

ID	2F_09.03.Zefiro-Europe.TRS.389
Description	Use of the emergency brake, for whatever reason, shall automatically cut all traction power, without the facility of reapplying traction power while the emergency brake is being applied.
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.828	Two cab to cab trainline for traction safe shall be provided.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.03.Zefiro-Europe.CONCEPT.909	The traction is enabling by energizing a trainline	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.1113	Traction safe trainline shall be de-energized when: C1) Green Loop (not all doors are closed) C2) No active cab C3) Brake pipe pressure less than 3,5 bars C4) Emergency Brake applied (Emergency Brake Loop De-energized) C5) TCMS DO are de-energized	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.1585	it shall be possible to de-energize any traction safe trainline using TCMS DO	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.1594	If TOWED mode is NOT active (from the active cab), TCMS shall de-energized both Traction safe trainline from the active cab (by means of safe DO) any time pressure in Brake Pipe drops below 350kPa. DO shall be reset with pressure riese over 400kPa Pressure signal for trigger shall be derived from second lowest valid TCG sensor in the train shall be used. (8 TCG sensor available in each train)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.1396	Pressure switch installed in DM1/DM8 shall measure the pressure in the brake pipe and de-energise the traction safe loop if brake pipe pressure is less than 3.5 bars.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.1401	If the brake pipe pressure switches are in different positions for more than 10 seconds an event shall be generated for the maintainer.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_09.03.Zefiro-Europe.TRS.334
Description	The traction demand shall be blocked to all motors in the train if any of following conditions is true: 1) master controller is in brake position 2) ATP service brake is active 3) too high speed in washing- or coupling mode OR 4) fault in speed calculation
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1052	Traction shall be blocked (traction block set) if any of the following conditions are set. (Traction block means that positive traction/brake reference is not allowed by TCMS) 1) Urgency brake is applied 2) Emergency brake is applied 3) No driver's desk is active 4) All Doors are not closed and locked from doors (including doors are released) 5) Master controller is in Brake Position 6) Master controller fault is detected 7) DBV in service brake position (if Direct Brake is enabled) 8) Low brake pipe pressure (brake SFRS concept.2174,1503) 9) Brake request from ATP is active 10) Inhibition traction is applied by the ATP 11) Passenger Emergency Brake is requested 12) The active cab is coupled 13) External power supply is connected at standstill 14) Traction safe loop is de-energised 15) No valid direction is set 16) Direction is changed at speed 17) Too low brake effort 18) Traction block request from washing speed mode 19) Traction block request from coupling speed mode 20) External air supply is connected (can be overridden from TOD) 21) Brake test is active 22) Traction test is active 23) Speed calculation fault 24) No traction is available 25) Emergency operation mode activated with master control handle in traction position 26) Change of driving mode with master control handle in traction position 27) No valid configuration 28) Decoupling sequence is active 29) Serious fault not acknowledged within 15sec 30) Change of direction with Master controller in Traction position 31) Traction block request from DIS system 32) ATP enabled in the not active cab (presence of life signal) 33) MCH backup mode switch active in DEP brake mode 34) - Speed limitation reset by TCMS	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1539	To re-establish traction, once traction-block conditions have been reset, the requested effort shall be first set to 0 (Master controller shall be put in coasting position or traction setpoint shall be put to 0 when "master controller back-up mode" is enabled). This is valid for all conditions exept when traction block is determined by ATP system at speed. In this case effort will be re-established as soon as the block is removed	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

2F_09.03.Zefiro-Europe.CONCEPT.1367	If the Traction safe trainline is considered to be energised the PCU shall be notified. Traction is considered to be enabled when below conditions is true: 1) both DI connected to a traction safe trainline are (1) on at least one trainline In case it is not possible to check conditon above (DI on a trainline are inchoerent or invalind) TCMS has to check that following condition are true: – all doors (external and cab) in the train are closed and locked – the brake pipe pressure is above 4.0 bar – a cab is active in the train – Emergency Brake is not applied (Emergency Brake Loop energized) The status of the Emergency operation switch and the Doors interlock bypass switch shall be considered for above conditions to avoid false indication of events.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1283	If traction is considered to be enabled according requirement 2F_09.03.Zefiro-Europe.CONCEPT.1367 and the traction safe trainline status sent from the PCU indicates that any Traction safe input is de-energised then an event shall be generated.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

TCMS shall check coherence between any of following elements and the related traction safe trainline:
- RBLE1/RBLE2 intervention in the active cab
- RTS1/RTS2 intervention in the active cab
- PST intervention in the first two motor cars in the leading half train (half train with active cab)
- TCMS DOs for Traction safe in the active cab

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1587	If emergency operation is NOT active and Green Loop is Energized (or bypassed): Any time relay RBLE1 is de-energized (Dig Input, 1 → 0), TCMS shall immediatly check that related PST (DM car switch) status is still closed (Dig Input = 0) and DO on TL1 are still both closed (1), so after 0,1s (parameter to be tuned during commissioning) TCMS shall check that : – RTS1 relay status is de-energized (Dig Input 0 → 1) – Related Traction Safe Trainline (TL1) shall be de-energized (feedback DI = 0) In case RTS1 or TL1 are not read as de-energized an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

2F_09.03.Zefiro-Europe.CONCEPT.1593	If emergency operation is NOT active and Green Loop is Energized (or bypassed): Any time relay RBLE2 is de-energized (Dig Input, 1 → 0), TCMS shall immediatly check that related PST (M car switch) status is still closed (Dig Input = 0) and DO on TL2 are still both closed (1), so after 0,1s (parameter to be tuned during commissioning) TCMS shall check that after : – RTS2 relay status is de-energized (Dig Input 0 → 1) – Related Traction Safe Trainline (TL2) shall be de-energized (feedback DI = 0) In case RTS2 or TL2 are not read as de-energized an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1588	Everytime emergency brake is released (RBLE1 energized in the active cab, Dig Input 0 → 1), TCMS shall check that as soon as TCG sensor in the DM car with active cab is above 450kPa, – Related PST switch shall be reset (Dig Input 1 → 0) – RTS1 shall be energized (Dig Input 1 → 0) within 0,5s otherwise an event shall be set The event shall not be triggered if MCB of RTS1 power supply is tripped.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1589	Everytime emergency brake is released (RBLE2 energized in the active cab, Dig Input 0 → 1), TCMS shall check that as soon as TCG sensor in the M car nearest to the active cab is above 450kPa, – Related PST switch shall be reset (Dig Input 1 → 0) – RTS2 shall be energized (Dig Input 1 → 0) within 0,5s otherwise an event shall be set The event shall not be triggered if MCB of RTS2 power supply is tripped.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1597	Everytime emergency brake is released (RBLE1 energized in the active cab, Dig Input 0 → 1), TCMS shall check that before PST pressure switchis in the DM car with active cab is trigger (1 → 0): – RTS1 is still de-energized (Dig Input 1) In case RST1 is energised (0) before PST is trigger an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1598	Everytime emergency brake is released (RBLE2 energized in the active cab, Dig Input 0 → 1), TCMS shall check that before PST pressure switch in the M car nearest to the active cab is trigger (1 → 0): – RTS2 is still de-energized (Dig Input 1) In case RST2 is energised (0) before PST is trigger an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

2F_09.03.Zefiro-Europe.CONCEPT.1590	If Green Loop is energized (or bypassed) and RTS1 is energized (DI = 0), as soon as DO on traction safe line 1 (TL1) are energized (0→1), TCMS shall check that – TL1 status shall trigger (0→1) whithin 0,5s otherwise an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1591	If Green Loop is energized (or bypassed) and RTS2 is energized (DI = 0), as soon as DO on traction safe line 2 (TL2) are energized (0→1), TCMS shall check that – - TL2 status shall trigger (0→1) whithin 0,5s otherwise an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1592	If there are NO local active cab in all consist of the train (including when parking mode is active), TCMS shall check that both Traction Safe trainline are de-energized (all feedback DI = 0) otherwise an event shall be set (including detection of faulty trainline)	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1347	If Emergency Brake loop is de-energized and the PCU indicates that any Traction safe input is energised an event shall be generated.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1285	If any door closed relay (RGL1, RGL2) are indicating that at least one door is open and the green loop is not bypassed then if the any PCU indicates that a Traction safe input is energised an event shall be generated.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1286	If the brake pipe pressure is below 2.3 bar and the PCU indicates that any Traction safe input is energised an event shall be generated.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1115	If traction block is active then an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.827	One input for Traction safe shall be provided on the MOCU/M to interrupt switching.	Derived Requirement	4S_05.-.Propulsion		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1287	The status of Traction safe input shall be forwarded to TCMS via the IP bus.	Derived Requirement	4S_05.-.Propulsion		N/A
2F_09.03.Zefiro-Europe.CONCEPT.829	Any traction demand or brake demand shall be blocked if the traction safe input is deenergised. The traction safe input shall affect both traction and dynamic brake.	Derived Requirement	4S_05.-.Propulsion		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1417	If both Traction safe loop are de-energized then Traction block shall be applied.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1179	Traction block shall be set if no traction is available at standstill.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1575	Traction block shall be set in case ATP is enabled in the not active cab (presence of life signal)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1576	In case of Traction block applied due to ATP enabled in the not active cab, an advice shall be displayed to the driver asking to exclude ATP in the other cab	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

3.1.5 Traction Safe Trainlines Bypass

3.1.5.1 *Function Design*

Traction safe can be inhibited or overridden in case of emergency to clear the track. This action has to be confirmed actively by the driver.

Traction safe can be inhibited by means of two sealed latched switches:
- can inhibit just the signal “Door interlocking bypass”;
- can bypass the whole Traction safe loop (both trainlines) by means of an "emergency operation switch".

The pressure switch "PST" located in leading motor cars will be used to cut traction in case of EB

3.1.5.1.1 Normal condition

Traction safe shall be overridden in case of malfunction to clear the track and start traction.

3.1.5.1.2 Failure cases

- Traction override is not working properly, and no traction is given
- Unwanted Override

3.1.5.1.3 Failure detection

- Done by the driver (even with the emergency operation active, traction is not available).
- Message on the TOD

3.1.5.1.4 Failure handling

- Train has to be towed
- Operational restriction and maintenance required

3.1.5.2 *Requirements*

ID	2F_09.03.Zefiro-Europe.TRS.343
Description	It shall be possible for the driver to manually override Traction safe in case of malfunction to clear the track and start traction.
Safety level	SIL 2

ID	2F_09.03.Zefiro-Europe.TRS.389
Description	Use of the emergency brake, for whatever reason, shall automatically cut all traction power, without the facility of reapplying traction power while the emergency brake is being applied.
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1054	Traction safe can be inhibited or overridden in case of emergency by rotating the dedicated two latched switches on the driver desk: By means of the '**Doors interlocking bypass**' the following reason for traction block shall be overridden: – All doors are not closed and locked from doors By means of the '**Emergency Operation Switch**' the following reason for traction block shall be overridden: 1) All reasons except emergency brake (via pressure switch PST) that opens the traction safe loop and including Traction safe loop de-energised.	Derived Requirement	4S_08.07.01.-.Drivers Desk		SIL 2

2F_09.03.Zefiro-Europe.CONCEPT.1117	Traction safe can be inhibited or overridden in case of emergency by pushing the dedicated two latched switches on the driver desk: By means of the '**Doors interlocking bypass**' the following reason for traction block shall be overridden: 1) All doors are not closed and locked from doors By means of the '**Emergency Operation Switch**' the following reason for traction block shall be overridden: – All reasons except emergency brake (via pressure switch PST) that opens the traction safe loop and including Traction safe loop de-energised.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.1118	Traction safe shall be inhibited or overridden in case of emergency by pushing the dedicated two latched switches on the driver desk: By means of the '**Doors interlocking bypass**' the following reason for traction block shall be overridden: – All doors are not closed and locked By means of the '**Emergency Operation Switch**' the following reason for traction block shall be overridden: 1) All reasons including Traction safe loop de-energised.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1055	The interruption of the door interlocking trainline and traction safe bypass shall be inhibited temporarily as long as a switch in the cab is activated.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.1056	A latched switches to bypass the door interlocking trainline shall be provided in the cab.	Derived Requirement	4S_08.07.01.-.Drivers Desk		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.1057	The operation of the door interlocking bypass function shall be supervised by TCMS, generating an event if the button in cab for door interlocking bypass is activated	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.845	The interruption of the traction safe trainline shall be inhibited temporarily as long as a button or switch in the cab is activated.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.846	A latched switches to bypass the traction safe trainline shall be provided in the cab.	Derived Requirement	4S_08.07.01.-.Drivers Desk		SIL 2

3.1.6 DNRA Alarm

3.1.6.1 *Function Design*

The DNRA alarm occurs when a non-rotating axle is detected.
Any axle is considered locked if the difference between reference speed (calculated) based on the speed signals of all four axles per car and the relevant axle speed (measured) is greater than 50 km/h + 0.3 * reference speed [km/h] for more than 10 s (UIC standards parameters).

DNRA supervision is performed by BCU and as redundancy by WSP-CU on the same time on car level.
If a non-rotating axle is detected, WSP and BCU interrupt the dedicated train line and communicate the alarm via safe protocol over IP.
The DNRA supervision is performed also in multiple composition; in this case DNRA alarm is activated on rear train, the TCMS master detect the alarm by DNRA trainline and by IP communication.

The DNRA is shown by an indication element "DNRA alarm" on driver's desk, if the alarm is active the indication element is steady illuminated and when the alarm occurs the buzzer is activated. In addition, the DNRA alarm is presented on the TOD.

In case of DNRA alarm a speed reduction is realized if the current speed is >80 km/h and then the speed limitation is applied.
Based on the design principle of the applied parking brake, in case the DNRA alarm TCMS applies immediately a direct brake application if the speed is above 90 km/h. The brakes are applied until the target speed of 80 km/h is reached. Beyond the target speed of 80 km/h the direct brake application is automatically released by TCMS to allow the driver to clear the track in a not safe situation at low speed (tunnel, etc.).
The TCMS automatic brake application is supervised by internal safe supervision logic. In case the target speed of 80 km/h is not reached in a specified time, TCMS applies the emergency brake to ensure the train is braked in safe condition.
The aim of the automatic brake function is to avoid a locked wheel set over a long time period.

The DNRA system is composed of:
- BCU: Brake control unit DNRA function per car
- WSP-CU: WSP control unit for redundant DNRA function
- SS1-4: Double speed sensor per axle (sensors with double channels one for BCU and the other for WSP-CU)
- DNRA train line feed from the non-active cab
- Safe communication over IP
- DNRA alarm indication lamp



In case of locked axle condition and detection the BCU and/or the WSP-CU energises the relay Rss1 and/or Rss2, to interrupt the train line and send a dedicated IP signal to TCMS.

In each DM car the DNRA train line is connected to two relay Rdnra1 and Rdnra2. Contacts of these two relays and the signals received over IP allow TCMS to control the "DNRA alarm" indication lamp located on the driver's desk and the activation of the audible alarm signal.
TCMS supervises the coherence between the alarm status received via IP bus and the DNRA trainline, the alarm setting conditions and TCMS actions are reported in the table “DNRA Handling” together the actions requested to the driver and the effects on the service.

TCMS supervises also the two DNRA relays Rss1 and Rss2 in each car.

In case of any failure in the DNRA system a maintenance event is set. During the complete brake test the output of the BCU and WSP-CU is set to de-energise the DNRA train line to test the complete signal transfer to the driver.

The TCMS is able to detect an undue de-energization of the loop “*DNRA*” because for every car of the vehicle it is available:
- the “*Blocked Axle*” information from every BCU and WSP via IP Safe bus

In this case only a maintenance event is generated.

Every time that not all the DNRA alarm sources not detect the same DNRA status an event for the maintenance is generated,

When the DNRA alarm occurs, the driving personnel stop the train as soon possible and, after an inspection, can under his own responsibility remove the speed limit at 80 km/h, utilizing the cabin sealed selector GC-S006. In this case, the DNRA function is bypassed and the maximum speed becomes equal to 200 km/h.

Case n.	loop	WSPx via bus (x= 1..8)	BCUx via bus (x= 1..8)	DNRA Alarm & Driver Event	Diagnostic Events	TCMS Actions	Driver Guide	Effects on the Operation
1	-	-	-	-	-			Nothing
2	x	x	X	 DNRA Alarm: Axle x Blocked	- DNRA Alarm from WSPx via IP - DNRA Alarm from BCUx via IP	- turn on DNRA lamp on the driver desk - buzzer activation for 15 seconds or until the acknowledgeon the TOD - speed reduction if > 80km/h - 80 km/h speed limitation	At standstill: - Inspection by short vehicle movements to check the rolling of the blocked axle reported on the TOD. - If the alarm is confirmed then the train requires maintenance, if possible free the line at low speed (20km/h) - if no anomaly is detected, isolate the axle via the RICF cock or the coach via the RIEP distributor, push the "Recognize DNRA" button on the bogie page to deactivate the event and the pictogram, resume the service - If the DNRA event occurs again then activate DNRA bypass via GG-S005	- Vehicle Stopping - If confirmed alarm by inspection free the line at low speed (20km/h)) - If undue alarm then resume the running without speed limitation - If DNRA bypass is necessary then the speed is limited to 200km/h
3	x	x	-	 DNRA Alarm: Axle x Blocked from WSP	- DNRA Alarm from WSPx via IP - DNRA IP signals not congruent from BCUx	- turn on DNRA lamp on the driver desk - buzzer activation for 15 seconds or until the acknowledgeon the TOD - speed reduction if > 80km/h - 80 km/h speed limitation	At standstill: - Inspection by short vehicle movements to check the rolling of the blocked axle reported on the TOD - If the alarm is confirmed then the train requires maintenance, if possible free the line at low speed (20km/h) - if no anomaly is detected then switch off the WSPx via GG-F00 3 where the alarm has been detected and resume the running without speed limitation	- Vehicle Stopping - If confirmed alarm by inspection free the line at low speed (20km/h)) - If undue alarm then resume the running without speed limitation
4	x	-	X	 DNRA Alarm: Axle x Blocked from BCUx	- DNRA Alarm from BCUx via IP - DNRA IP signals not congruent from WSPx	- turn on DNRA lamp on the driver desk - buzzer activation for 15 seconds or until the acknowledgeon the TOD - speed reduction if > 80km/h - 80 km/h speed limitation	At standstill: - Inspection by short vehicle movements to check the rolling of the blocked axle reported on the TOD - If the alarm is confirmed then the train requires maintenance, if possible free the line at low speed (20km/h) - if no anomaly is detected then switch off the BCUx via GG-F00 6 where the alarm has been detected and resume the running without speed limitation	- Vehicle Stopping - If confirmed alarm by inspection free the line at low speed (20km/h)) - If undue alarm then resume the running without speed limitation -
5	-	x	X	 DNRA Alarm: Axle x Blocked	- DNRA Alarm from BCUx via IP - DNRA Alarm from WSPx via IP - DNRA signals not congruent from loop	- turn on DNRA lamp on the driver desk - buzzer activation for 15 seconds or until the acknowledgeon the TOD - speed reduction if > 80km/h 80 km/h speed limitation - 80 km/h speed limitation	At standstill: - Inspection by short vehicle movements to check the rolling of the blocked axle reported on the TOD - If the alarm is confirmed then the train requires maintenance, if possible free the line at low speed (20km/h) - if no anomaly is detected, isolate the axle via the RICF cock or the coach via the RIEP distributor, push the "Recognize DNRA" button on the bogie page to deactivate the event and the pictogram, resume the service If the DNRA event occurs again then activate DNRA bypass via GG-S005	- Vehicle Stopping - If confirmed alarm by inspection free the line at low speed (20km/h)) - If undue alarm then resume the running without speed limitation - If DNRA bypass is necessary then the speed is limited to 200km/h
6	x	-g	-	-	- DNRA trainline Undue De-energization	-	Nothing	Nothing

7	-	-	X	 DNRA Alarm: Axle x Blocked from BCU	- DNRA Alarm from BCUx via IP - DNRA IP signals not congruent from WSPx - DNRA signals not congruent from loop	- turn on DNRA lamp on the driver desk - buzzer activation for 15 seconds or until the acknowledgeon the TOD - speed reduction if > 80km/h - 80 km/h speed limitation	At standstill: - Inspection by short vehicle movements to check the rolling of the blocked axle reported on the TOD - If the alarm is confirmed then the train requires maintenance, if possible free the line at low speed (20km/h) - if no anomaly is detected then switch off the BCUx where the alarm has been detected via GG-F006 and resume the running without speed limitation	- Vehicle Stopping - If confirmed alarm by inspection free the line at low speed (20km/h)) - If undue alarm then resume the running without speed limitation
8	-	X	-	 DNRA Alarm: Axle x Blocked from WSP arm: Axle x Blocked from BCU	- DNRA Alarm from WSPx via IP - DNRA IP signals not congruent from BCUx - DNRA signals not congruent from loop	- turn on DNRA lamp on the driver desk - buzzer activation for 15 seconds or until the acknowledgeon the TOD - speed reduction if > 80km/h - 80 km/h speed limitation	At standstill: - Inspection by short vehicle movements to check the rolling of the blocked axle reported on the TOD - If the alarm is confirmed then the train requires maintenance, if possible free the line at low speed (20km/h) - if no anomaly is detected then switch off the BCUx where the alarm has been detected via GG-F003 and resume the running without speed limitation	- Vehicle Stopping - If confirmed alarm by inspection free the line at low speed (20km/h)) - If undue alarm then resume the running without speed limitation •
9	x	-	-	 DNRA Alarm: Axle Blocked in carx	- WSPx or BCUx DNRA not available or IP communication failure	- turn on DNRA lamp on the driver desk - buzzer activation for 15 seconds or until the acknowledgeon the TOD - speed reduction if > 80km/h - 80 km/h speed limitation	At standstill: - Inspection by short vehicle movements to check the rolling of the axle in the car x reported on the TOD - If the alarm is confirmed then the train requires maintenance, if possible free the line at low speed (20km/h) - if no anomaly is detected then resume the running without speed limitation f the DNRA event occurs again then activate DNRA bypass via GG-S005	- Vehicle Stopping - If confirmed alarm by inspection free the line at low speed (20km/h)) - If undue alarm then resume the running without speed limitation - If DNRA bypass is necessary then the speed is limited to 200km/h

Table 2 – *DNRA Handling*

3.1.6.1.1 Normal condition

In normal conditions the DNRA function is supervising that the axles are not locked for more than 10s during running speed > 50 km/h.

3.1.6.1.2 Failure cases

1. DNRA not detected from BCU
2. DNRA not detected from WSP
3. DNRA train line remains energised on DNRA request
4. DNRA train line de-energised without order (DNRA not detected from BCU and from WSP)

3.1.6.1.3 Failure detection

1. Event indication on TOD of DNRA signals not congruent
2. Event indication on TOD of DNRA signals not congruent
3. Event indication on TOD of DNRA signals not congruent
4. Event indication on TOD of DNRA signals not congruent

3.1.6.1.4 Failure handling

1. In case of BCU detection failure, the DNRA function is redundant by WSP-CU.
2. In case of WSP-CU detection failure the DNRA function is redundant by BCU.
3. In case of DNRA train line remains energised on DNRA request, WSP and BCU send the DNRA alarm over IP.
4. In case of DNRA train line is de-energised without order, TCMS detects the DNRA status from WSP and BCU over IP, no driver action. No effect on train level. An event is created for maintenance.

3.1.6.2 *Requirements*

ID	2F_05.02.Zefiro-Europe.TRS.48
Description	DNRA system shall supervise each axle separately. When DNRA detects a wheel set non-rotating an alarm shall be presented to the driver.
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.HR13	A train line for DNRA indication connected to BCU and WSP-CU relay contacts shall be provided. The train line shall be fed from the non-active rear cab of trainset. The DNRA train line shall be connected via relay contacts Rdnra1 and Rdnra2 to 2 TCMS safe digital input in each DM-car.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2

2F_09.03.Zefiro-Europe.CONCEPT.HR14	The DNRA Alarm indication lamp shall be active in the active cab If one of the following DNRA Alarms occurs: - DNRA Alarm: Axle x Blocked - DNRA Alarm: Axle x Blocked from WSP(also on remote WSP in multiple unit configuration) - DNRA Alarm: Axle x Blocked from BCU(also on remote BCU in multiple unit configuration) DNRA Alarm: Axle Blocked in carx	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.HR15	A DNRA alarm lamp connected to a TCMS safe output shall be provided on driver's desk in the cab. The DNRA lamp shall be visible to driver.	Derived Requirement	4S_08.07.01.-.Drivers Desk	HW HMI	SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.HR16	*If RDNRA1 or RDNRA2 digital input is low in the DM car with cab active the DNRA loop is considered de-energized*	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.HR17	**GIVEN** (("*Request DNRA*" on safe IP by WSPx is invalid **OR** DNRA function is not Available by WSPx) **OR** ("*Request DNRA*" on safe IP by BCUx is invalid **OR** DNRA function is not Available by BCUx)) **WHEN** ("*DNRA Loop*" de-energized) **THEN** (TCMS shall set " *DNRA Alarm: Axle Blocked in carx* " C1A Serious Fault)	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.HR18	IF ("Request DNRA" on safe IP by BCUx is active AND “Request DNRA" on safe IP by WSPx-CU is NOT active) **THEN** (TCMS shall set " *DNRA Alarm: Axle x Blocked from BCU* " C1A Serious Fault)	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.HR19	IF ("*Request DNRA*" on safe IP by WSPx-CU is active AND "Request DNRA" on safe IP by BCUx is NOT active) **THEN** (TCMS shall set " *DNRA Alarm: Axle x Blocked from WSP* " C1A Serious Fault)	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.HR20	IF (WSPx and BCUx signal over IP indicate *DNRA Alarm)* THEN TCMS shall set " *DNRA alarm: Axle x Blocked* " C1A Serious Fault	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.HR21	If both control units BCU and WSP-CU set IP signal "DIA DNRA function disturbed" a C2 serious fault shall be set on IDU that DNRA function is not available in this car and the train speed shall be reduced to 200 km/h by the driver.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.HR24	If a DNRA Alarm occurs the acoustical alarm TCMS buzzer shall sound for 5s. During DNRA test test the buzzer shall be not sound.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.HR25	The BCU and WSP-CU DNRA ouput relay contacts Rss1 and Rss2 shall be connected to TCMS input in each car.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.HR26	The DNRA train line relay contact Rdnra1 and Rdnra2 shall be connected to TCMS safe inputs when local cab is enabled..	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.HR27	If the Rss1, Rss2 status contacts and the relay contact of the DNRA train line Rdnra1 and Rdnra2 input in TCMS have a different status an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

2F_09.03.Zefiro-Europe.CONCEPT.HR101	In case of multiple unit, the TCMS shall check the Rss1, Rss2 status contacts of each train with status of relay DNRA train line Rdnra1 and Rdnra2 located in the DM-car with enabled cab in case of different status an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.HR28	TCMS shall verify the plausibilty of the relay contacts Rdnra1 and Rdnra2, in case they are not the same status an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.HR29	The BCU DNRA output shall be connected to a relay Rss1, the relay contact shall open the DNRA train line. The WSP-CU DNRA output shall be connected to a relay Rss2, the relay contact shall open the DNRA train line.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.HR30	A contact of Rss1 and Rss2 shall be connected in series to a TCMS input for relay supervision.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.HR31	If the BCU and WSP-CU DNRA relay RSS TCMS input is set (1) and the BCU and WSP-CU IP signal "DNRA request active" is not align an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.HR32	**GIVEN** (TCMS active) **WHEN** ((TCMS detects by digital input "*DNRA request active*" in a certain car) **AND** (BCU IP signal "*DNRA request active*" is not active in the same car) **AND** ("DNRA test" is not active)) **THEN** (TCMS shall set "*DNRA request from BCU incoherency*" event in the car)	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.HR33	**GIVEN** (TCMS active) **WHEN** (("*DNRA request from BCU incoherency*" event is active in a certain car) **AND** ((Vehicle comes out from standstill) ***OR*** ("*DNRA Alarm*" pictogram on IDU is switched-off acknowledging it by "Safe *Soft Key*"))) **THEN** (TCMS shall reset "*DNRA request from BCU incoherency*" event in the car)	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

2F_09.03.Zefiro-Europe.CONCEPT.HR34	**GIVEN** (TCMS active) **WHEN** ((TCMS detects by digital input "*DNRA request active*" in a certain car) **AND** (WSP-CU IP signal "*DNRA request active*" is not active in the same car) **AND** ("DNRA test" is not active)) **THEN** (TCMS shall set "*DNRA request from WSP-CU incoherency*" event in the car)	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.HR35	**GIVEN** (TCMS active) **WHEN** (("*DNRA request from WSP-CU incoherency*" event is active in a certain car) **AND** ((Veichle comes out from standstill) **OR** ("*DNRA Alarm*" pictogram on IDU is switched-off acknowledging it by "*Soft Key*"))) **THEN** (TCMS shall reset *"DNRA request from WSP-CU incoherency*" event in the car)	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.HR36	In case of a locked axle detection, the IP signal "DNRA request active" have to be received from TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.HR37	In case of a DNRA request active IP signal "DNRA request active" reported by BCU/WSP-CU a maintenance event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.HR38	In case of a DNRA Alarm detection a pictogram like to DNRA lamp shall be displayed on "driving page"	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.HR39	The DNRA pictogram pictogram on the TOD be active in the active cab when one of the following DNRA Alarms occurs(also on remote train): : - DNRA Alarm: Axle x Blocked - DNRA Alarm: Axle x Blocked from WSP - DNRA Alarm: Axle x Blocked from BCU - DNRA Alarm: Axle Blocked in carx	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

2F_09.03.Zefiro-Europe.CONCEPT.HR40	If DNRA train line is de-energized (relay contacts Rdnra1 or Rdnra 2 are reset (0)) AND no IP signal "DNRA request active" is set either by BCU or WSP-CU an event shall be set indicating that the DNRA trainline is de-energized without order.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.HR41	If the IP signal "DNRA request active" is set by BCUx AND the IP signal "DNRA request active" is not set by WSPx-CU Then an event shall be set indicating “DNRA IP signals not congruent from WSPx”	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.HR42	If the IP signal "DNRA request active" is set by WSPx-CU AND the IP signal "DNRA request active" is not set by BCUx Then an event shall be set indicating “DNRA IP signals not congruent from BCUx”	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.HR43	**GIVEN** ((NO Valid "*Request DNRA*" is active on safe IP by WSP in each consist of trainset) **AND** (NO Valid "*Request DNRA*" is active on safe IP by BCU in each consist of trainset)) **WHEN** ("*DNRA Loop*" is de-energized) **THEN** (TCMS shall set "*DNRA Undue De-energization*" maintenance event)	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.HR44	**GIVEN** (A valid "*Request DNRA*" is active on safe IP by WSPx) **OR** (A valid "*Request DNRA*" is active on safe IP by BCUx) **AND** ("*DNRA - Loop*" is energized) **THEN** (TCMS shall set "*DNRA* signal not congruent from loop” maintanance event)	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.HR53	**GIVEN** ("DNRA Alarm: Axle x Blocked” OR “DNRA Alarm: Axle x Blocked from WSP” OR “DNRA Alarm: Axle x Blocked from BCU” OR “DNRA Alarm: Axle Blocked in carx” C1A Serious Fault is active) **WHEN** (Vehicle at "*Standstill*") AND ("*Safe Key*" on IDU pushed in the active cab) AND All the alarm setting conditions are not present) **THEN** (All "*DNRA Alarms*" C1A Serious Fault are reset)	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.HR45	**WHEN** the DNRA serious fault has been reset **THEN** The DNRA pictogram on IDU shall be reset	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.HR46	**WHEN** the DNRA serious fault has been reset **THEN**	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2

	the DNRA lamp and the DNRA alarms event shall be reset				

ID	2F_05.02.Zefiro-Europe.TRS.342
Description	In case of undue intervention of "DNRA - Alarm" , driver shall be able to by-pass speed limitation.
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.HR47	A DNRA by pass selector is present on the Driver Desk	Derived Requirement	4S_09.02.09.-.Conventional Train Control – Circuits		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.HR48	The DNRA bypass selector status is connected to TSMS by 2 safe Digital Inputs	Derived Requirement	4S_09.02.09.-.Conventional Train Control – Circuits		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.HR49	In case DNRA by-pass switch of the enabled cab will be activated TCMS should: – inhibit Full Service Brake demand due to DNRA alarm; – set the speed limitation value to 200 kmph	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.HR50	In case DNRA by-pass switch of the enabled cab will be activated TCMS should: - Disable DNRA detection in each consist of trainset; - at standstill switches-off DNRA lamp, if yet active after pushed the safe key ,	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.HR51	In case DNRA by-pass switch of the enabled cab will be activated an event should be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.HR52	In case DNRA by-pass switch of the enabled cab will be activated TCMS should: - inhibit deceleration supervision function of the full service brake in case of DNRA request only	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2

3.1.7 Train Speed Reduction / Train stopping due to Alarms

3.1.7.1 *Design*

If the following alarms occur(also in multiple unit composition):
- Parking brake applied at speed
- DNRA Alarm C1A serious Fault
- HABD Alarm C1A serious Fault

A Max service braking shall be automatically actuated until reaching the speed of 80 km/h.
TCMS implements a deceleration supervision, checking that after 8 sec from the braking request the average deceleration in the last 3 sec and then until reaching 80km/h is <= 0.4m/sec2. The 5 second delay to verify the deceleration value is compliance with the jerk protection and the brake cylinders pressure rise times.
If the calculated deceleration is higher than the target, emergency braking is applied.

3.1.7.2 *Requirements*

2F_09.03.Zefiro-Europe.CONCEPT.HR22	If a DNRA alarm occurs TCMS shall activate full direct service brake independent of service brake mode if the speed ≥ 90 km/h. If a DNRA alarma is active the traction limit shall be set at a speed of 80 km/h.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

	If the train speed is < 80 km/h the full service brake request shall be reset and the brake application shall be released.				
2F_09.03.Zefiro-Europe.CONCEPT.HR58	If a HABD C1A serious Fault detected by thermal fuse or by PT100 occurs TCMS shall activate full direct service brake independent of service brake mode if the speed is ≥ 90 km/h. If the HABD alarm is active the traction limit shall be set at a speed of 80 km/h. If the train speed is < 80 km/h the full service brake request shall be reset and the brake application shall be released.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.HR59	If the parking brake is reported applied at speed by BCU IP signal "Parking brake applied at speed v > 3 km/h" TCMS shall activate full direct service brake independent of service brake mode if the speed ≥ 90 km/h. If the Parking brake alarm is active a traction limit shall be set at a speed of 80 km/h. If the train speed is < 80 km/h OR the parking brake applied by BCU and TCMS input is reset the full service brake request shall be reset and the brake application shall be released.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.HR23	TCMS shall supervise the deceleration time of the full direct service brake application for different function: – Parking brake applied at speed – DNRA Alarm C1A serious Fault – HABD Alarm C1A serious Fault Based on the initial speed and the linear deceleration value of 0.4 m/s² to reach the target speed of 80 km/h the supervision shall be performed. The deceleration supervision shall be activated 8 seconds after the service braking application in the last 3 sec window and then until reaching 80km/h. If the deceleration is out of the range TCMS shall apply urgency brake until the target speed < 80 km/h.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2

3.1.8 Emergency Operation mode

3.1.8.1 *Function Design*

In case of emergency, for example for fire incident or failure of electrical equipment, the train can be operated in emergency operation mode by selecting the "emergency operation mode" sealed switch.

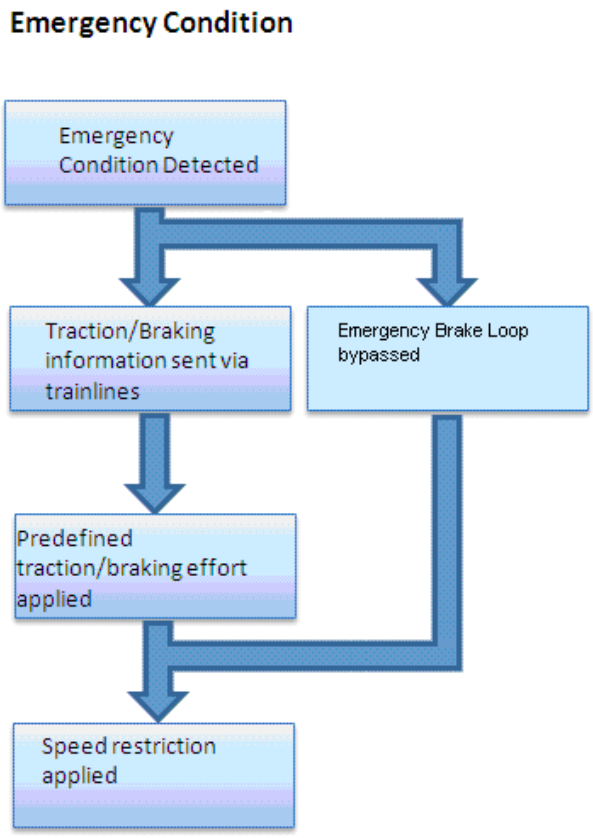
A dedicated trainline is provided to inform all the involved system

Conditions that involves Emergency Operation:
1) TCMS cannot read the controls in the cab (e.g. a fire incident that isolate Ethernet connection)
2) Electrical failure involving Emergency Brake loop (e.g. trailine interruption, failure on equipment commanding EB via electrical loop)

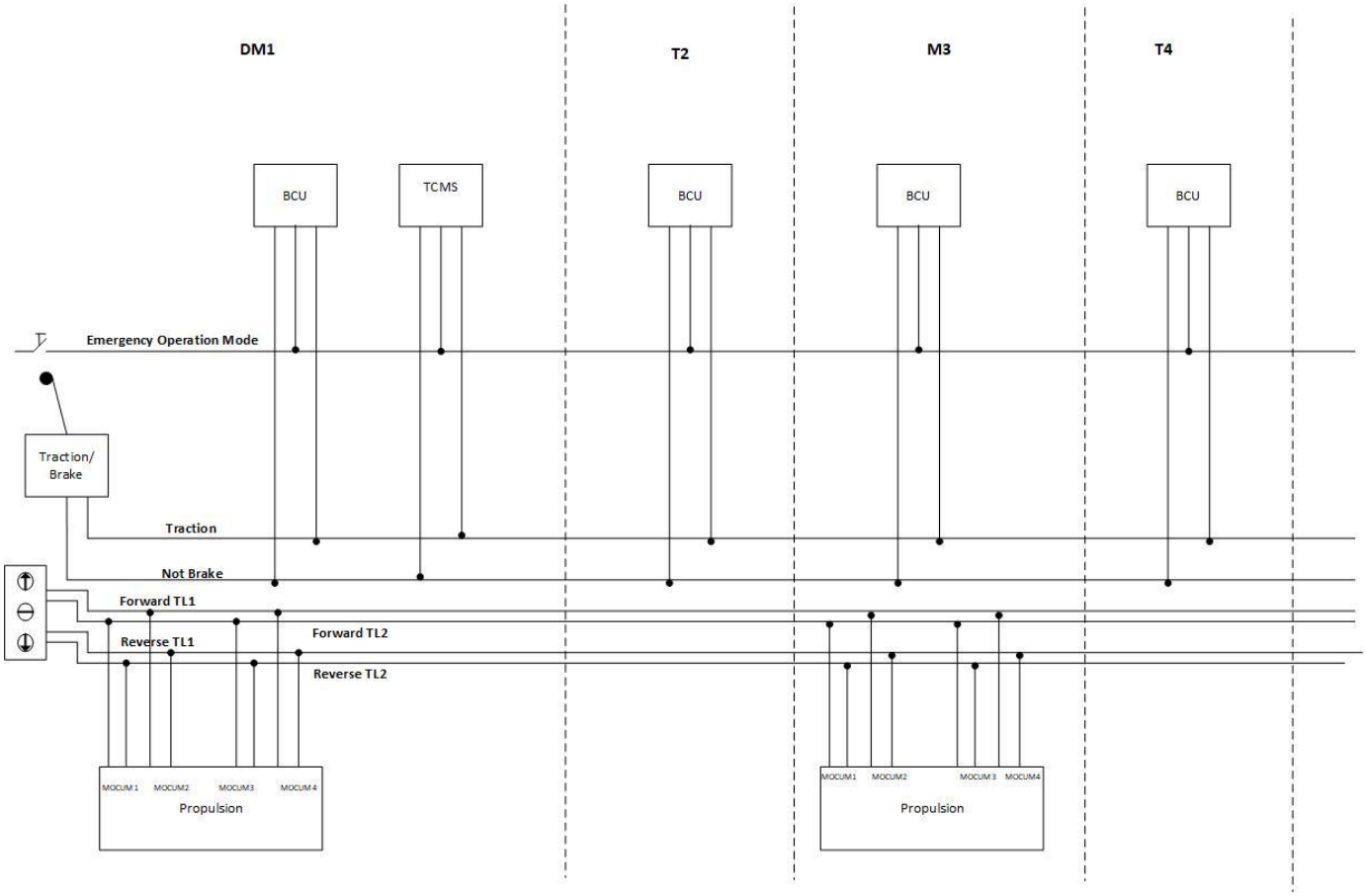
When the "Emergency Operation" mode is active:
1) the traction/brake information will be distributed by means of 2 fire resistant trainlines that connect directly the master controller with all the BCUs and TCMS. The demand will be forwarded by the TCMS to the propulsion system via the IP bus
2) the emergency brake loop will be bypassed (energizing directly the two EVAPP valves by means of the Emergency Operation trainline.

The safe application of an Emergency brake is guaranteed by the direct pneumatic venting of the Brake pipe via EB Pushbuttons and DBV,.

- The traction effort required is pre-defined in order to keep a train speed of 80km/h



Architecture of emergency traction/brake trainlines



3.1.8.1.1 Normal condition

The emergency operation mode shall be activated anytime by the driver using a sealed switch. The manual activation shall ensure that the driver is aware about the risk and the limitation that this mode brings. In addition, the speed limit set by TCMS should avoid dangerous situations due to high speed
In emergency operation mode, the traction and braking demand from the brake handle to TCMS may not be available.
.
If the emergency operation mode' is activated, then the 'emergency operation mode' trainline is energised. The status of the 'emergency operation mode' trainline shall be monitored by TCMS in every 4 car segment.
If the 'emergency operation mode' trainline is set then a predefined amount of traction and brake demand shall be demanded of the BCU and MCMs depending on if the traction or brake trainlines are energised. The traction and brake trainlines are read by the BCU in each car.
In addition, in case of failure to the emergency brake loop the "emergency mode" trainline is used to keep EVAPP valves (that are used to vent the brake pipe) energized allowing the train to move and escape from dangerous situations.
'Forward' and 'reverse' trainlines functionality shall not be affected.

3.1.8.1.2 Failure cases

- No Traction/Braking effort is manageable through the trainlines
- Some traction/braking effort signals are not available
- Emergency Brake Loop cannot be bypassed

3.1.8.1.3 Failure detection

- Done by the driver (no possibility to move the train)
- Done by the driver (some master controller positions do not provide any response)
- Done by the driver (train cannot move and Brake pipe pressure is 0bar)

3.1.8.1.4 Failure handling

1) Tow is needed
2) Proceed with the available efforts
3) Bypass EVAPP valves manually (using RIVAPP cocks)

3.1.8.2 *Requirements*

ID	2F_09.03.Zefiro-Europe.TRS.337
Description	The distribution of traction demand level shall be ensured to at least 50% of all MOCUM along the train for minimum 15 minutes in case of fire.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.812	If Emergency operation mode is active and the Emergency traction demand input is set then the traction demand shall be regulated to keep a constant train speed of 80km/h. The traction demand sent from the Master controller shall be ignored.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_09.03.Zefiro-Europe.TRS.196
Description	The driver shall confirm the emergency operation by a dedicated button or switch. Emergency operation shall be active only if this switch is activated by the driver.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1046	There shall be a 'emergency operation mode' sealed switch on the drivers desk.	Derived Requirement	4S_08.07.01.-.Drivers Desk		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1120	The 'emergency operation mode' sealed switch on the drivers energises the 'emergency operation mode trainline'.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1119	The status of the 'Emergency operation mode' trainline shall be connected to TCMS	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1121	The status of the 'Emergency operation mode' sealed switch shall be connected to TCMS	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1047	When the 'emergency operation mode' sealed switch is pressed there shall be an event logged	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1449	When "emergency operation" mode is active, and indication shall be displayed on TOD. In addition the prescribed speed limitation has to be shown	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1122	If there is a discrepency between the 'emergency operation mode' sealed switch and the 'emergency operation mode' trainline then an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

ID	2F_09.03.Zefiro-Europe.TRS.231
Description	In case of an incident, the activation of braking operation shall not cause braking down to stand still. Any evacuation drive must be ensured with minmum time delay.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL

2F_09.03.Zefiro-Europe.CONCEPT.524	In case of any brake application due to fault the brake application shall release after switch over to emergency operation when MCH is moved into traction position.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1410	If MCH is in traction position (traction trainline energized before emergency operation trainline) then traction shall be blocked until traction trainline is reset.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_09.03.Zefiro-Europe.TRS.192
Description	The emergency operation mode is used as backup for traction and brake control in case of a major incident like fire.
Safety level	N/A

ID	2F_09.03.Zefiro-Europe.TRS.194
Description	The master controller shall be connected directly to this backup communication path though being independent. In emergency operation only the traction/brake controller and the indirect brake lever (DBV) can be used. The emergency operation mode shall allow the train to run even in case of fault on the Emergency Brake Loop.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.394	The master controller shall provide a separate contact set to be connected to **2** trainlines.	Derived Requirement	4S_08.07.01.-.Drivers Desk		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1049	The 'traction trainline' and 'brake trainline' shall be set by the traction and brake areas of the Master controller.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1124	The 'traction trainline' and 'brake trainline' shall be supervised by TCMS	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1126	If the 'emergency operation mode trailine' is energised then the TCMS will react according to the change of status of the trainline.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.656	The traction demand also in Emergency Mode operation shall be sent to propulsion control via IP bus.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1611	When Emergency Operation mode is activated with Doors released, if communication with driver's desk is ok, once traction is demanded an advisory shall be set to the driver indicating that a closing command of the doors shall be issued.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1420	When traction trainline is set during emergency operation mode then HV shall be reactivated if HV was active when emergency mode was activated.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1609	In case of automatic re-configuration of HV due to missing communication with driver's desk, when the driver puts the MCH in braking position for a time less than 2 seconds and then move it to coast TCMS shall open the LCB and lower the pantograph. When MCH is moved again to traction position TCMS shall re-configures automatically HV according previous configuration.	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_09.03.Zefiro-Europe.CONCEPT.1610	In case driver's desk key position, LCB switch status, pantograph selectors or line voltage selector cannot be detected or redundant TCMS feedbacks are incoherent an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1447	If emergency operation mode is activate, the Emergency Brake loop shall be bypassed energizing directly EVAPP valves	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 3
2F_09.03.Zefiro-Europe.CONCEPT.1448	If an Emergency Brake is commanded via Master Controller (MCH), during emergency operation mode, the EVAPP valves bypass shall be released	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 3

Emergency Operation shall be able to move the train, also in case of multiple fire alarms set in technical areas of both half trains.

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1599	In case of Fire alarm detected in technical areas, if Emergency Operation is activated, following actions shall be granted: 1) all pantographs shall be available to the driver, except for those ones previously faulty (not in case of ADD intervenction by-pass) OR cut-out; 2) LCB of the raised pantograph shall be allowed to be closed; 3) ACMs of the half train with pantograph raised shall be allowed to starts; 4) MCMs of the half train with pantograph raised shall be allowed to starts; **Note:** 220V sockets and MV auxiliary contactors (Gallley in DM1, Bistro in M3) shall be kept deactivated. Also, separation contaction of MV (contactor 6) and HV roof line disconnector have to kept open.	Derived Requirement	4S_09.03.05.-.TCMS Software		

3.1.9 Train Speed Calculation

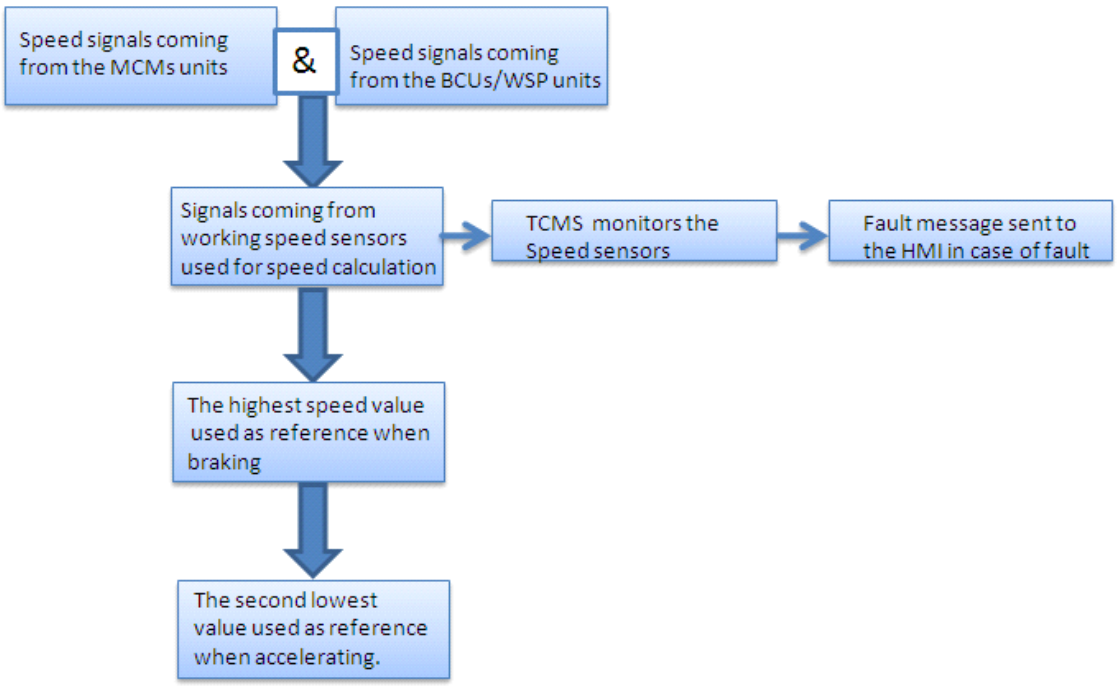
3.1.9.1 *Design*

Purpose of this function is to describe how to calculate train speed. Train speed is calculated by means of the available speed sensors belonging to the brake system and the propulsion system.
All speed sensors correctly working will be used for the train speed calculation but for the following function:

1) doors interlocking:
   BCU/WSP will be used to provide the reference speed.

– visualization to the Driver in case ATP cut-off:
  WSP speed will be displayed in a safe way
  (optional DIS will be used - non safe)

Speed Calculation:



3.1.9.1.1 Normal condition

There is consistency between the different speeds signals coming from the 8 car unit used for train speed calculation.

3.1.9.1.2 Failure cases

- Speed signal of one or more sensor/system are not align to train speed but at least 1/3 of the speed sensors in the 8 car unit used for train speed calculation are correctly working.
- Less than 1/3 of the speed sensors in the 8 car unit used for train speed calculation are correctly working

3.1.9.1.3 Failure detection

1) TOD → Message sent to identify where are located the faulty speed sensor
2) TCMS → Major alarm activated on the TOD & Diagnostic A-event buzzer to be activated

3.1.9.1.4 Failure handling

1) TCMS Excludes the faulty speed sensor/s from reference speed. Maintenance needed at the end of the service.
2) Try a vehicle master reset, if the faults are still present then proceed to the closest station and retire the vehicle from the service. Maintenance needed.

3.1.9.2 *Requirements*

ID	2F_09.03.Zefiro-Europe.TRS.199
Description	The speed signals from all available BCUand WSP on IP bus shall be used for train speed calculation.
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.973	Each BCU/WSP (per car) and MCMs will send the speed signal per axle to TCMS and TCMS shall calculate the train speed based on valid speed sensors.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

2F_09.03.Zefiro-Europe.CONCEPT.1130	MCMs shall send the speed signal per axle to the TCMS based on valid speed sensors.	Derived Requirement	4S_05.-.Propulsion		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1131	Each WSP shall send the speed signal per axle to the TCMS based on valid speed sensors.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.HRI 02	Each BCU shall send the speed signal per axle to the TCMS based on valid speed sensors.	Derived Requirement	4S_07.-.Braking System		SIL 0

ID	2F_09.03.Zefiro-Europe.TRS.158
Description	The speed sensors from brake- and propulsion system in a 8 car unit will be used to calculate the train speed. The strategy of the calcualtion is to use the hightest speed value when braking and coasting and the second lowest value when accelerating.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.974	TCMS shall calculate the speed on a 8 car unit basis to ensure fast signal sampling times. The speed sensor from the first 8 car unit (considering as first car the active cab both in single train and in multiple operation) will be used to calculate the train speed according to the following strategy: 1) The highest speed value shall be used as reference value when braking, 2) The second highest value of trailer cars shall be used as reference value when not at braking. 3) If no valid trailer car speed is present then the lowest speed value shall be used when not at braking.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1414	If no valid speed can be calculated traction block shall be applied.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.HR105	In Multiple unit , TCMS of slave train shall sent its calculated speed to master train	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_09.03.Zefiro-Europe.TRS.161
Description	The speed sensors shall be continually supervised and faulty sensors shall be excluded from the speed calculation
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.975	The speed signals shall be continually supervised. An event shall be sent and presented on the TOD which speed sensor is faulty. Faulty speed signals shall be excluded for generating train speed	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.962	When the difference between max and min speed is > 6km/h an event shall be generated. The comparison shall not be active during slide	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.963	For axles with two speed sensors (motor axles) the two speeds shall be compared and both be excluded from the speed calculation if the speeds differs more than max of (1% of train speed and 1km/h). The comparison shall not be active during slide	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.399	The axle speed from all MOCUM axles shall be provided to TCMS.	Derived Requirement	4S_05.-.Propulsion		N/A
2F_09.03.Zefiro-Europe.CONCEPT.400	The axle speed from all BCU/WSP axles shall be provided to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.860	TCMS shall send the train speed to propulsion control for slip regulation. The signal shall be calibrated considering the wheel diameters.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_09.03.Zefiro-Europe.TRS.200

Description	The speed calulation for – visualisation in the cab – doors interlocking – DSD function shall be provided by ATP or other SIL2 computer, e.g. the Brake Controller BCU or WSP. In case of ATP cut out : – TCMS shall calculate the speed by means of the available WSP. –
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1450	If the ATP speed signal is not available (ATP cut off) the speed from the WSP shall be displayed on the TOD. (optionally DIS speed signal will be used)	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0

3.1.10 Speed Limitation

3.1.10.1 *Design*

3.1.10.1.1 Normal condition

The train speed shall be limited depending on the train status.

3.1.10.1.2 Failure cases

- Speed limitation is not applied when a restriction is required
- Speed Limitation applied when not requested

3.1.10.1.3 Failure detection

- Detected by driver
- Detected by the driver by means of message on the TOD

3.1.10.1.4 Failure handling

1) Speed to be regulated by driver
2) Try a vehicle master reset, if the faults are still present then proceed to the closest station and retire the vehicle from the service. Maintenance needed.

3.1.10.2 *Requirements*

ID	2F_09.03.Zefiro-Europe.TRS.223
Description	If ATP is cut out, the speed shall be limited to 80km/h (TBD).
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.431	Traction block shall be applied when DIS request traction block. (For example ATP cut out)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_09.03.Zefiro-Europe.TRS.155
Description	Depending on the status of the train speed limits are applied. The most restrictive speed limit shall be used. 1) coupling speed (1 km/h) 2) washing speed (5 km/h) 3) Reverse running direction (40 km/h) 4) ice scraping (manual speed reduction by driver(250 km/h) The following speed limits are defined in the following degraded conditions: 1) emergency operation mode (80 km/h) 2) driving without ATP (ATP cut out) (50 km/h) 3) DNRA bypass (200 km/h) 4) HABD local bypass (200 km/h) 5) Both Pantograph Control unit in fault (160 km/h) 6) running moving the cover couper (50 km/h) 7) running with coupler cover open (250 km/h) The following speed limits are defined for Alarm activation: 1) secondary suspension fault (220 km/h) 2) Major mechanical fault (manual speed reduction by driver) (250 km/h) 3) IMS alarm (220km/h) 4) DNRA alarm (80 km/h) 5) HAB Alarm (80 km/h) 6) parking brake applied (80 km/h) 7) pre Alarm for bearings or gearbox high temperatures, manual speed reduction by driver (200km/h) If these conditions occur in multiple units configuration on remote Train, TCMS shall apply the most restrictive speed limit
Safety level	N/A

The speed limitations will be enforced by a combination of:
1) an automatic speed restriction by TCMS with no SIL and
2) a manual restriction done by the drivers following operational rules to achieve SIL2

The driver will notice that a speed limitation is in force:
- ATP isolated: The dedicated lamp (LCEA) on the driver desk is switched on and both ATP DMI are black (Traction block requested by DIS)
- coupling speed will be selected actively by the driver from standstill
- washing speed will be selected actively by the driver from standstill
-
- speed reductions due to IMS uses trainlines and alarm lamps in the drivers desk is described in the bogies SFRS.
- overheat or power limitations related to propulsion and high voltage will be detected internally in the propulsion system and the torque will be limited. I.e. the system protects itself without driver interaction.
- Pantograph error: see the line voltage SFRS.

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.236	The speed limit with the lowest (most restrictive) value in the trainset shall always be used to limit the speed automatically by reducing the traction demand when the speed limit is reached. The traction demand shall be ramped down to zero at a speed of 10% beyond the speed limit. This shall ensure that the train still can drive with fulload at an inclination.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1451	The active speed limit shall be displayed on the TOD	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1579	Whenever a speed limitation is reset by TCMS, a traction block shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.1.11 Speed Calibration

3.1.11.1 *Design*

All axle speeds used for Speed Calculation shall be calibrated using an independent speed source (ATP)to determine the axle diameters. The calibration shall be executed from TCMS and PCUs, then TCMS corrects the BCU axle speeds, the PCUs correct the Propulsion axle speeds with the correct calibration factor calculated during the calibration phase.

3.1.11.1.1 Normal condition

The calibration procedure shall start upon manual request from the driver and shall use ATP speed as reference. The train shall be coasting to enable the calibration. The diameters calculated for all speed signals received by TCMS shall be stored in TCMS

3.1.11.1.2 Failure cases

- Calibration fails to initiate when calibration required
- Calibration initiates but no update of wheel diameter performed

3.1.11.1.3 Failure detection

- Uneven regulation of speed and uneven ride
- Uneven regulation of speed and uneven ride

3.1.11.1.4 Failure handling

- Maintenance needed before the train is entered into traffic after a wheel set change
- Maintenance needed before the train is entered into traffic after a wheel set change

3.1.11.2 *Requirements*

ID	2F_09.03.Zefiro-Europe.TRS.154
Description	The axle speed signals from the BCU/WSP shall represent a rotational speed only.
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.407	The PCUs (MOCUM) shall provide speed signals for all motor axles to the TCMS.	Derived Requirement	4S_05.-.Propulsion		N/A
2F_09.03.Zefiro-Europe.CONCEPT.408	The WSP controllers shall provide speed signals for all axles to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.HRI 01	The BCU controllers shall provide speed signals for all axles to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 0

ID	2F_09.03.Zefiro-Europe.TRS.201
Description	The wheel diameters, calculated during the calibration phase, shall be stored in the TCMS and used for correction of the speed signals received from BCU. The calibration shall be done using the speed from the ATP system.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1368	The speed from ATP shall be sent to all PCU's in the train to be used for wheel diameter calibration,.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.HR03	A calibration command shall be sent to all PCU's in the train to execute the wheel diameter calibration,	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.554	The wheel diameters shall be calibrated towards the reference speed when command for wheel diameter calibration is set from TCMS.	Derived Requirement	4S_05.-.Propulsion		N/A

2F_09.03.Zefiro-Europe.CONCEPT.1400	The signal provided by the ATP shall be used to verify the train speed measured by TCMS.	Derived Requirement	4S_09.06.02.-.National Safety System		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.HR106	In case of multiple unit configuration, the slave train shall command speed calibration using as reference the speed of master train	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_09.03.Zefiro-Europe.TRS.212
Description	Speed calibration shall ensure the consideration of wheel diameters in the train speed signal calculation.
Safety level	N/A

ID	2F_09.03.Zefiro-Europe.TRS.203
Description	The calibration procedure shall only be activated manually from a driver or maintainer after a workshop visit affecting the wheel diameters.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.404	A speed calibration request shall be activated if the condition belowe are soddisfy, between a speed range of 100-300 km/h: 1) Calculated Train Speed from TCMS, derived from BCU/MOCUM, differs more than 3km/h from ATP speed, when train is in coasting or in traction and no slip/slide is active from BCU/MOCUM with speed higher than 100 km/h for more than 10 seconds. TCMS shoud store it and if the condition 4 to 9 are valid a calibration requerst has to be performed. This condition has to be reset if the calibration is passed **OR** 2) The difference between max and min speed read by all the sources is > 6km/h, when train is in coasting or in traction and no slip/slide is active from BCU/MOCUM with speed higher than 100 km/h for more than 10 seconds. TCMS shoud store it and if the condition 4 to 9 are valid a calibration requerst has to be performed. This condition has to be reset if the calibration is passed **OR** 3) If one speed of the MOCUM is geater than 3 km/h compared with the train speed. This criteria is valid only if train speed is more than 100 km/h for 10 seconds and no slip/slide is active from the PCU. If this condition is active the TCMS should store it and if the condition 4 to 9 are valid a calibration requerst has to be performed. This condition has to be reset if the calibration is passed If condition 1 **OR** 2 **OR** 3 is active, the out put has to be put in **AND** with the conditions below 4) Pneumatic brake not applied for more than 5 seconds **AND** 5) Traction/brake effort reference is between 0 and +20% for more than 5 seconds **AND** 6) One LCB closed in the train (this condition has to be the same in multiple conposition, one LCB closed per train) **AND** 7) ATP Acceleration/Decceleration < 0.10 m/s^2 **AND** 8) The speed range is between 100-300 km/h, the ATP signal has to valid **AND** 9) No slip/slide is active for more than 5 seconds	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

	If a speed calibration is activated, an event should be set in Telediagnostica, there should be also an event in Telediagnostica that explaince why the calibration is requested, (condirion 1, 2 or 3). The duration of the calibration should be at least 15 seconds The TCMS should activate also an event in Telediagnostica if the calibration is passed or not 1) The calibration is passed if the condition from 4 to 9 have been present for at least 15 seconds 2) The calibration is failed if the condition from 4 to 9 have been present for less than 15 seconds. The event should explain why the calibration has failed				
2F_09.03.Zefiro-Europe.CONCEPT.88	During speed calibration the BCU speeds shall be calibrated towards the ATP reference speed.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.405	If ATP speed signal is not available the speed calibration cannot be initiated.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.406	An event shall be set in case a speed calibration request is activated specifying which is the cause that has triggered the request	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.411	The button to trigger the speed calibration on TOD should not be foreseen	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A

3.1.12 Distance Counter

3.1.12.1 *Design*

The travelled distance function is an internal TCMS software function.

3.1.12.1.1 Normal condition

A counter shall increase by 1 every 1 kilometer and stored in non volatile memory

3.1.12.1.2 Failure cases

Counter not increasing when train has been in traffic

3.1.12.1.3 Failure detection

No increase in kilometers travelled from last upload of event data

3.1.12.1.4 Failure handling

Maintenance needed

3.1.12.2 *Requirements*

ID	2F_09.03.Zefiro-Europe.TRS.163
Description	An odometer function logging the travelled distance shall be provided for maintenance use. The data shall be stored in non volatile memory and be available for the Telediagnostica application.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.94	An odometer function shall logg the travelled distance and store it in battery backed up memory. The data shall be available for the Telediagnostica application.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.93	The resolution of the Odometer function shall be 1km	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

3.1.13 Master Reset

3.1.13.1 *Design*

Master reset is a TCMS software controlled reset of certain permanent failures and blockings.
Different systems use interlocks for blocking of faulty equipment. The fault interlock must have a reset function with unlimited number of available resets that is available for the maintenance/service people when train is in Maintenance mode only.
In case of multiple unit configuration, the Master reset is sent also to remote Train.

3.1.13.1.1 Normal condition

The master reset can be performed by the maintenance personnel during maintenance or by the driver during operation from the TOD. Master reset is allowed by the TCMS just at standstill condition (v=0).

3.1.13.1.2 Failure cases

Master reset is demanded but no reset is performed.

3.1.13.1.3 Failure detection

Faulty equipment not reset and continue to influence train performance

3.1.13.1.4 Failure handling

Not known (Manually disconnect/reconnect the power supply of the faulty unit to force the faulty system to be re-started in order to verify if in this way the permanent block can be eliminated.)

3.1.13.2 *Requirements*

ID	2F_09.03.Zefiro-Europe.TRS.165
Description	Different systems use interlocks for blocking the use of faulty equipment. That interlock has a reset function that is available for the driver/maintener. The reset is called “Master Reset”.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.99	A button on TOD shall be provided to trigger a "master reset".	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A

ID	2F_09.03.Zefiro-Europe.TRS.213
Description	A "Reset" function shall be provided for the maintenance crew to reset applied interlocks of faulty equipment. The reset accessible to maintainers shall be unlimited but be protected against unauthorised access.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.432	A "master reset" shall be sent to all subsystems when following conditions are met: 1) all pantograph are lowered; 2) no external power supply is connected; 3) standstill condition	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1613	A dedicated event shall be set to the Driver when Master reset PB is pressed but any of the following conditions is not met: – all pantograph are lowered; – no external power supply is connected; – train not standstill	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.238	The "Master Reset" signal shall be received from TOD indipendently from which mode is active (Driver, Maintenance)	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A

2F_09.03.Zefiro-Europe.CONCEPT.1135	The "Master Reset" function can be unlimited requested by Driver and Maintainer only in the active cab.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
~~2F_09.03.Zefiro-Europe.CONCEPT.1390~~	~~The "Master Reset" function is allowed by the TCMS only at standstill condition (v=0).~~	~~Derived Requirement~~	~~4S_09.03.05.-.TCMS Software~~		~~SIL 0~~
2F_09.03.Zefiro-Europe.CONCEPT.HR108	The "Master Reset" command in multiple composition shall be sent to remote train	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.605	An event shall be logged if "master reset" has been trigged	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

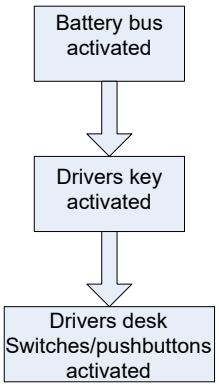
3.1.14 Drivers Desk supervision

3.1.14.1 *Design*

If the cab is activated (occupied), the indicators and control elements are active.
They will be hardwired activated by switch/relay when battery bus and drivers key is activated.

Lamp test uses a local pushbutton. All control lamps, gauges in control of hardware or software are tested. The hardwired lamps will be tested hardwired.

Block diagram Drivers desk activation ½ trainset



3.1.14.1.1 Normal condition

The driver's desk indicators and controls are available when the cab is activated.
A lamp test function is available by pressing a pushbutton on the driver desk

3.1.14.1.2 Failure cases

- Specific button, lamp or display not working
- Lamp test is not working

3.1.14.1.3 Failure detection

- Driver observation
- Driver observation

3.1.14.1.4 Failure handling

- Depends on indication or control button failure
- Depends on indication or control button failure

3.1.14.2 *Requirements*

ID	2F_09.03.Zefiro-Europe.TRS.168
Description	Control from driver desk buttons and switches shall only be active in the active cab (excluding the emergency stop pushbutton and the driver key which are active in all cabs).
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.104	All control elements in the cab besides the two emergency stop push buttons, the battery on push button, the parking mode pushbutton and the drivers key shall be disabled in the non-active cab. The supply to those controls shall be disconnected.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.608	All control elements in the cab besides the two emergency stop push buttons, the battery on push button, the parking mode pushbutton and the drivers key shall be disabled in the non-active cab.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1577	Events shall be set for main driver actions on driver desk (eg LCB command, panto, doors, key, handles, bypass switches etc..)	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

ID	2F_09.03.Zefiro-Europe.TRS.169
Description	Indicators, lamps and buzzers controlled by hardware directly, shall be disabled in the non-active cab a part from the paking mode lamp and door release indication.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.245	Indicators, lamps and buzzers controlled by TCMS software shall be disabled in the non-active cab a part from the paking mode lamp.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.434	Indicators, lamps and buzzers controlled by hardware directly, shall be disabled in the non-active cab a part from the paking mode lamp and door release indication.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

ID	2F_09.03.Zefiro-Europe.TRS.171
Description	Test of hardware controlled lamps, indicators shall be done upon manual command to ensure the correct functionality. (SIL 2 by means of the perfomance of the manual test by the driver -> driver´s manual)
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.436	A lamp test button shall be provided on the drivers desk.	Derived Requirement	4S_08.07.01.-.Drivers Desk	HW HMI	N/A
2F_09.03.Zefiro-Europe.CONCEPT.501	The lamp test for hardware controlled lamps shall be triggered directly from the lamp test button. The lamp test button shall also be connected to TCMS digital input.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.03.Zefiro-Europe.CONCEPT.377	Lamp test of hardware controlled indicators (gauges, lamps) shall be done by manual command from the activated desk.	Derived Requirement	4S_08.07.01.-.Drivers Desk		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1136	If the lamp test button is pressed, all indicators shall be tested: 1. the lamps shall be completely lit	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

ID	2F_09.03.Zefiro-Europe.TRS.214
Description	Test of TCMS controlled lamps, indicators shall be done upon command to ensure the correct functionality.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.251	Lamp test of TCMS controlled indicators (gauges, lamps) shall be done by manual command from the activated desk. If the lamp test button is pressed, all indicators shall be tested: – the lamps shall be completely lit – the TOD (the primary and the secondary) shall show a white screen testing all pixles	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.502	When the lamp test button is pressed all indicators shall be activated simultaneously, the software controlled and the hardware controlled buttons. The synchronisation of all indicators shall be ensured.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

3.1.15 Serious Fault Indication

3.1.15.1 *Design*

A "serious fault" shall be notified to the driver via a dedicated indication. If a fault occurs that needs to be monitored or that implies a speed reduction, this lamp shall be lit.
When an event occurs, the driver shall acknowledge it, otherwise Traction block will be set until the acknowledge is done

There are different types of fault that can be classified as "serious":

- C1A Events in which the TCMS sets automatic braking in order to to reduce the speed up to 80 km/h and applies traction cut out and speed limitation to 80 km/h;

- C1B Events in which the TCMS sets traction cut out and speed limitation to 80 km/h;

- C2 Events in which the TCMS applies traction cut out.

When another desk is enabled the indication will be switched on again and the driver shall recognize the fault before start (if possible)
The value of ΔTmax should be no more than 24hrs
The "serious fault" indication can be reset only after that the proper actions are performed (by the driver of by maintenance operators).
In any case if the maximum amount of time is elapsed and the serious fault is still active the lamp will be lit again and TCMS will not allow the train to start.
If this occurs during the normal service operations next time the train is stopped with released doors it will be not allowed to move again.
Alarms or faults that involve an automatic reaction from the train (eg emergency brake application) are not classified as "serious faults"
A dedicated redundant LED shall be installed on the driver desk

The Serious fault indication lamp shall be directly controlled by the TCMS via digital output. Because of this the lamp behaviour can be tuned according to the requests of the different authority.

3.1.15.1.1 Normal condition

In normal condition both LEDS shall be activated after TCMS identify serious fault.

3.1.15.1.2 Failure cases

- Only one of the two LEDs is activated after major fault identification
- None of the controls is activated after major fault identification
- One or both indicators are active even if there is not fault.

3.1.15.1.3 Failure detection

1. Done by the driver
2. Done by the driver due to the additional information on the TOD
3. Mute the Buzzer by means of the dedicated acknowledge button and/or ignore the Lamp.

3.1.15.1.4 Failure handling

- During the maintenance done to eliminate the major fault, also the LED needs to be repaired
- During the maintenance done to eliminate the major fault, also the LEDs need to be repaired
- Maintenance is needed at the end of the service.

3.1.15.2 *Requirements*

ID	2F_09.03.Zefiro-Europe.TRS.226
Description	A "serious fault" shall be indicated to the driver via a dedicated lamp on the drivers desk.
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1085	A redundant lamp with two LEDs shall be provided on the driver desk.	Derived Requirement	4S_08.07.01.-.Drivers Desk		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.1142	A "serious fault" redundant LEDs shall be connected to TCMS output. The two LEDs shall be connected to 4 different RIO-S outputs on two independent RIO-S units.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2

ID	2F_09.03.Zefiro-Europe.TRS.376
Description	TCMS shall illuminate the "Serious fault" lamp on the driver's desk if a 'serious fault' has occured.
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1515	When a "serious fault" occurs the driver shall acknowledge it via softdkey on TOD. If no ack is detected within 15 seconds Tracion block will be set until the acknowledge is done.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1516	Serious fault alarm can be acknowledged using the safe key on TOD	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1550	When new "serious fault" is present the lamp on the driver's desk shall be lit and the buzzer in the cab shall sound The buzzer shall be muted when the ack is pressed	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1079	The lamp shall extinguish if conditions for rectification are fulfilled, i.e. the fault is bypassed, or fault condition is inactive.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.1540	If a second serious fault occurs when the lamp is already switched on, the buzzer shall start to sound and the new fault shall be acknowledged by the driver.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1519	When a serious fault is active and a new cab is activated, the lamp shall be switched on in the new cab and the fault needs to be again acknowledged by the driver. The same requirement shall be applied in multiple unit configuration	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.1581	Serious fault supervision shall be enabled after train start up as soon as following conditions are met: – One cab activated in train AND Pressure in Main reservoir is greater than 700kPa (from at least one BCU in each consist of the composition, for multiple unit operations)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2

	OR 1 - Cab is active for 10 minutes OR – Cab is active in train AND train is moving In case Automatic Brake test is active Serious fault supervision shall be disabled				
2F_09.03.Zefiro-Europe.CONCEPT.1522	Following events shall be considered as Serious Faults (according classes C1A, C1B, C2) C1A: – DNRA Alarm – HABD (Axle box) Alarm – HABD (Gearbox) Alarm C1B: – Axle box over temperature – Gearbox over temperature C2: – Brake pipe neutral and isolation valve (EVNEU/EVISO) supervision – Triple Stage Supervision (missing stage application) – Triple Stage Supervision (undue stage application) – DNRA function loss in one or more cars (DNRA disturbed) – Dragging Brake (undue brake application) – BCU and WSP fault (or missing comunication) in a single car – EB application valve and pressure switch supervision – RIO Debug Mode	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.1526	C1A class faults shall lit the lamp as long as proper actions are not performed and failure is not solved	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.HR110	In multiple unit configuration the serious fault conditions shall be checked also on slave train	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.1527	C1B class faults shall lit the lamp as long as the train is over the prescribed speed limit	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.1525	In case of C1B faults the related speed limit shall be displayed on TOD	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1572	C2 class faults shall be displayed (lamp is switched-on, alarm is shown and acknowledge is required)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL2
2F_09.03.Zefiro-Europe.CONCEPT.1528	For C2 class faults the lamp shall be switched off soon after the driver acknowledges the alarm	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.1517	A time counter shall be set as soon as an event classified as serious fault occurs (only for failures belonging to categories C1A, C1B and C2). The value of the counter [ΔTmax] shall be settable indipendently for each fault	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1521	The counter shall be reset as soon as the serious fault has been solved.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1518	If a counter is elapsed an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1424	The serious fault can be acknowledged once to temporarily continue driving. After the timer is elasped, next time the train is standing with released doors and the serious fault is still active, an alarm shall be displayed on TOD stating that the maintenance time is elapsed and it is not safe to drive the train.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

2F_09.03.Zefiro-Europe.CONCEPT.1425	The reason for the serious fault shall be stored in NVRAM for Diagnostic.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1426	An event shall be set when serious fault acknowledge button is pressed.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1427	The serious fault lamp shall be part of the lamp test.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1520	The serious fault lamp shall be part of the lamp test.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1595	Following condition shall be used to "isolated" related Serious Faults (C2 faults) once triggered BCU and WSP fault → RIEP/DIST cock on isolated position in the affected car Dragging brake → n/a (trigger condition will be reset automatically after driver isolation action) Triple Stage Supervision (undue stage application) → RIEP/DIST cock on isolated position in the affected car (in this case also the "missing stage supervision" fault will be isolated) EB application valve and pressure switch supervision (EVAPP) supervision → RIVAPP cock on isolated position for the affected valve	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2

3.1.16 TCMS configuration waiting indication

3.1.16.1 *Design*

Any change in the train configuration needs to be notified to the driver to avoid that wrong operation are performed while the Train Logic is busy
A lamp controlled by TCMS will be installed on the driver's desk. The activation of this lamp is done in such a way that if there is any problem and the TCMS is not responding the indication is lit
The lamp shall be steady lit if:

- a test of HV/Propulsion equipment is in progress
- HV disconnectors reconfiguration
- selected HV configuration is not in line with the catenary voltage
- TCMS is not responding and/or faulty

The lamp is flashing in multiple train operation if:

1) TCN is faulty
2) Bus Inauguration is not successful

3.1.16.1.1 Normal conditions

In normal condition the lamp should be off.
If the lamp is lit steady or flashing the driver shall wait for the TCMS to complete the ongoing operations

3.1.16.1.2 Failure cases

1) Lamp is not lit when ordered
2) Lamp is lit also if there isn't any fault

3.1.16.1.3 Failure detection

1) Done by the driver during lamp test

2) Done by the driver checking detailed information on TOD

3.1.16.2 *Requirements*

ID	2F_09.03.Zefiro-Europe.TRS.382
Description	One lamp indicating that the train configuration is changing shall be installed on Driver's Desk
Safety level	

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1487	One lamp indicating that the train configuration is in progress shall be installed on Driver's Desk	Derived Requirement	4S_08.07.01.-.Drivers Desk	HW HMI	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1488	The lamp shall be connected to a TCMS Digital Output in such a way that if the DO is low the lamp will be switched on	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1497	The lamp is connected to a TCMS Digital Output in such a way that if the DO is low the lamp is switched on The lamp shall be steadly lit if: – a test of HV/Propulsion equipment is in progress – HV disconnectors reconfiguration – selected HV configuration is not in line with the catenary voltage – TCMS is not responding and/or faulty The lamp is flashing in multiple train operation if: – TCN is faulty – Bus Inauguration is not succesful	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1508	If the train is waiting for the new configuration and the lamp is lit (steady or flashing) the software traction block shall be applied (if possible) and traction effort shall be set to zero	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.1.17 Cab doors supervision

3.1.17.1 *Design*

The cab doors will be equipped with two double contacts switch for indicating the status of the cab door. This is included in the door green loop circuit. The status signals are sent to TCMS. The green loop prevents the train from having traction without "all doors closed".
The cab door also has a double contact if door is cut out. This is also supervised by connection to TCMS.
The cab doors are simple passive doors only used by the train staff, they have no door control unit. A double contact can be connected with the "green loop" all doors closed and locked. In the Traction Safe loop is the green loop shown. The cab doors is included in the "green loop" line.
The information on the status of the double contacts has to be sent also to the TCMS

3.1.17.1.1 Normal condition

The green loop line opens when one door in train opens.

3.1.17.1.2 Failure cases

- Door is open without opening green loop
- Door is closed but green loop remains deenergised

3.1.17.1.3 Failure detection

- TCMS compares signals from door double contacts and the green loop. If they do not have the same status a failure shall be detected, and an event shall be set.
- All cab doors closed but no traction is possible

3.1.17.1.4 Failure handling

- Isolate the door
- Doors closed and locked bypass to be activated.

Cab door are manually operated.

3.1.17.2 *Requirements*

ID	2F_09.03.Zefiro-Europe.TRS.236
Description	If a cab door is open the green loop shall be interrupted
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.536	The closed contacts of the cab doors shall be inlcluded in the "all doors closed and locked (green) loop" .	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.1092	Auxiliary contacts used to inform about the status of the cab doors shall be connected to the TCMS by means of digital Input.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1289	An event shall be set in case the cab door is not closed	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1093	The status of the cab doors shall be displayed on the TOD(also in Multiple Unit).	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A

3.1.18 Driving Mode

3.1.18.1 *Requirements*

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1547	Entering "Driving Mode" shall be possible only if the following condition are fulfilled: – battery voltage is activated, AND – pantograph is up, AND – main circuit breaker is closed, AND – line voltage is available, AND – 1 cab is active	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_09.03.Zefiro-Europe.CONCEPT.1549	Normal driving mode shall be ended when: – battery voltage is switched off, OR – line voltage is not available, OR – 1 cab is deactivated	Derived Requirement	4S_09.03.05.-.TCMS Software		

3.1.19 Train Off Mode

3.1.19.1 *Requirements*

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1548	Train off mode shall be reached from driving mode by the following actions: – by pressing the 'Main circuit breaker off' button – by pressing the 'Pantograph down' button – by deactivating a cab with the key switch – by pressing the "Battery OFF" push button From train off mode the switch-over to driving mode or workshop mode is possible.	Derived Requirement	4S_09.03.05.-.TCMS Software		

3.1.20 Towing activation

3.1.20.1 *Design*

Towing mode affects the towed vehicle only, the towing vehicle is not affected by the towing except that the brake control must be done via the UIC pipe by the DBV controller. The braking function in towing mode is defined in the Braking SFRS.

3.1.20.1.1 Normal condition

The train that is about to be towed is coupled mechanically with locomotive or with other train of same type. The cab is then activated in the train to be towed and the Towing switch is then activated. If conditions for "towed" mode is fulfilled the train will be in "towed" mode. In "towed" mode following functions will be blocked in the towed train:

- lifting of pantograph
- closing of LCB
- traction
- dynamic braking

The HVAC and Interior lighting functionality available in "towed" mode will be dependent on the available 3-phase supply through the back-up braking function in the Propulsion system. See HVAC SFSR and Interior lighting SFRS for more information.

3.1.20.1.2 Failure cases

- Towing switch is faulty. The train will not be set in "towed" mode. The train must be towed as "dead" train according to operation regulations.
- "Towed" mode falsely set due to faulty Towing switch. The brake pipe supervision will not open the traction safe loop if the DBV controller is used to brake the train.

3.1.20.1.3 Failure detection

- An event will be indicated to the driver on the TOD if "towed" mode is activated. The absence of this event will be an indication to the driver that towing is not possible.
- The Towing switch will be supervised in all inactive cabs and an event will be indicated to the driver if towing switch is activated in an inactive cab.

3.1.20.1.4 Failure handling

- Towing must be done according to operation regulations. Maintenance required when train is coming to depot.
- Braking of the train must not be done by the DBV controller. Master controller shall be used instead. This will be managed by the driver according to operation manual. Maintenance required when train is coming to depot.

3.1.20.2 *Requirements*

ID	2F_09.03.Zefiro-Europe.TRS.215
Description	The towed V300 Zefiro shall be put into "towed" mode: the cab shall be activated and a dedicated switch for towing shall be activated.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1236	There shall be a five stable position switch on the driver's desk to activate towing: Position 1 - Towing/Pushed Position 2 - Towing/Pulled Position 3 - Towing not active Position 4 - Towed/Pulled Position 5 - Towed/Pushed 1 and 2 are "Towing" positions 4 and 5 are "Towed" positions	Derived Requirement	4S_08.07.01.-.Drivers Desk		N/A

					
2F_09.03.Zefiro-Europe.CONCEPT.1237	All the positions of the towing switch shall be notified to TCMS	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1248	If the Towing switch is in towed position then an event shall be set to indicate that the towing switch is in towed position.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1249	If the Towing switch is in towed position in a de-activated cab then an event shall be displayed on the TOD.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

ID	2F_09.03.Zefiro-Europe.TRS.217
Description	The switching into "towed" mode shall be interlocked until mechanical coupling is confirmed from the coupler sensor.
Safety level	N/A

ID	2F_09.03.Zefiro-Europe.TRS.216
Description	The following restrictions apply in towed mode: – brake control via the UIC brake pipe (IP braking), i.e. mechanical brake only. – No traction, line circuit breaker open – no dynamic braking – three phase supply only available from backup braking of the propulsion system (if possible).
Safety level	N/A

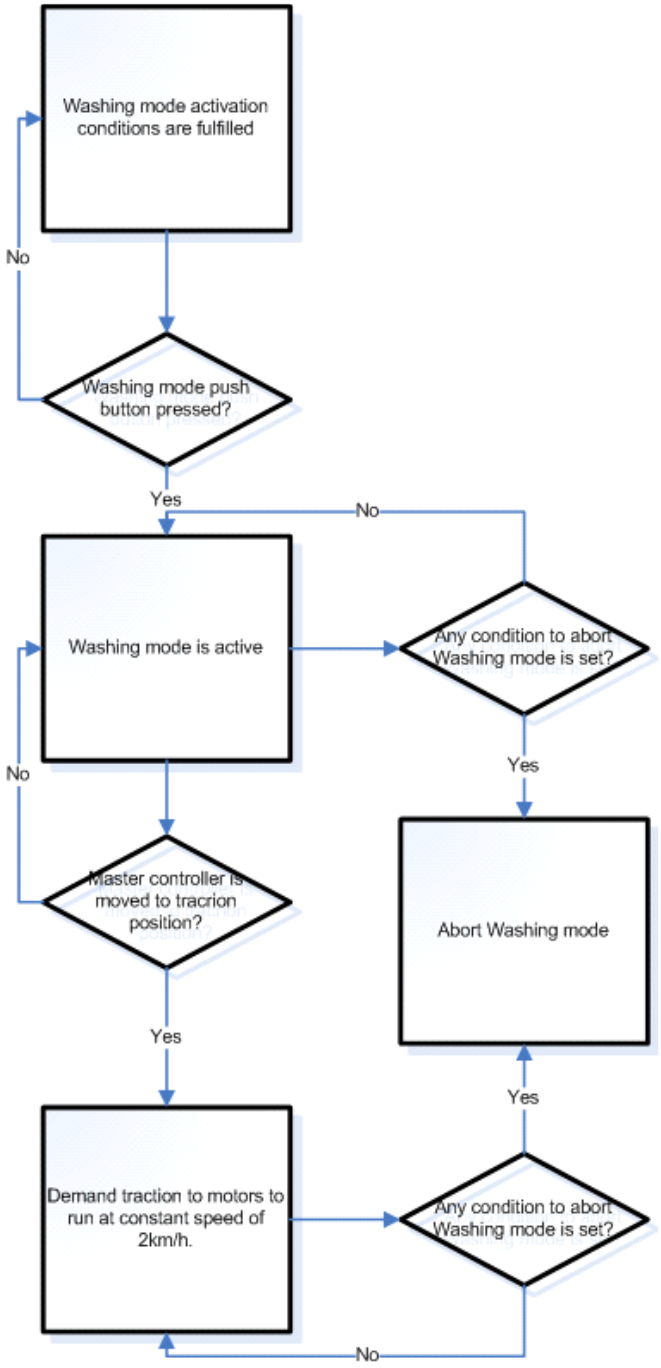
Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.453	The towed V300 Zefiro shall be put into "towed" mode and an event shall be displayed on the TOD to confirm that "towed" mode is set if - the cab is activated and - the Towing switch is in towed position and - line circuit breaker open and - pantograph down - valid direction is set (train not in neutral) If Towing mode switch is selected in: Treno soccorso (C) Treno soccorso (T)	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

	 TCMS shall close load shed contactors (BC2 and BC3)				
2F_09.03.Zefiro-Europe.CONCEPT.1580	When towing mode is activated a pictogram shall be shown on TOD	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A
2F_09.03.Zefiro-Europe.CONCEPT.457	If Towing switch is turned to towed position and any of following conditions is set then an event shall be set to the driver indicating that towing is not possible: – the cab is not active – line circuit breaker is closed – pantograph is up – direction is not set or not valid	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1238	If "towed" mode is active then all PCU's in the towed train shall be notified to enable back-up braking (charge DC-link for 3-phase supply).	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1239	If "towed" mode is active then the Master controller inputs shall be ignored.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1240	If "towed" mode is active then lifting of any pantograph in the towed train shall be blocked.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1241	If "towed" mode is active then closing of any LCB in the towed train shall be blocked.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1244	If "towed" mode is active then the traction/brake demand to the PCUs in the towed train shall be set to 0.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.526	If towing switch is in towed position and the LCBs are opened and the pantograph is down then the pressure switches on UIC brake pipe in traction safe loop shall be bypassed to allow back up braking of MCM.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL2

3.1.21 Washing speed mode

3.1.21.1 *Design*

Washing mode is used to get a slow steady speed while the train is being washed.
Below flowchart shows how Washing mode is selected/aborted and the washing speed control.



3.1.21.1.1 Normal condition

If conditions for Washing mode are set then it is possible for the user to activate the Washing mode by pushing a button on the TOD. If Washing mode is successfully activated then the user moves the master controller to any traction position. The train will then start to move at constant speed of 5km/h (could be changed up to 5km/h at runtime). The

movement of the train and Washing mode will be halted when the master controller is moved to neutral- or brake position.

3.1.21.1.2 Failure cases

- The speed limitation is not applied even if the washing mode has been selected. Movement through washing plant will be faster than desired.
- Washing mode is not activated by pressing the dedicated button. Movement through washing plant is not recommended since the train must be driving as normal which could cause the train to move faster than desired through the washing plant.
- Washing mode is activated without pressing the dedicated button. Train speed will be limited to 2km/h.

3.1.21.1.3 Failure detection

- Done by the driver/maintainer. Traction block willl be applied and an event will be set.
- Done by the driver/maintainer via TOD. The washing mode will not be indicated as active on the TOD.
- Done by the driver/maintainer (Speed reduction noted)

3.1.21.1.4 Failure handling

- Do not perform the washing procedure, maintenance needed
- Do not perform the washing procedure, maintenance needed
- If the train is not in service proceed with maintenance, if the train is in service tow could be needed.

3.1.21.2 *Requirements*

ID	2F_09.03.Zefiro-Europe.TRS.218
Description	The Washing mode shall be activated from the TOD.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.285	The TOD shall be equipped with a virtual push button to select the washing mode.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1140	When the Washing mode is active it shall be indicated on the TOD.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A

ID	2F_09.03.Zefiro-Europe.TRS.82
Description	Washing speed control shall only be possible to selected when: – the Master Controller is in position coast AND – train speed is 0 AND – traction is possible
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.462	Washing mode shall be activated if the washing mode virtual push button is pressed and the following conditions are fulfilled: – the cab is active AND – line voltage is active – the Master Controller is in neutral position (coast) AND – train speed is 0 (standstill) AND – traction is possible (not blocked) – DEP brake available and selected (if not washing speed mode is not allowed)	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1104	If washing mode virtual push button is pressed and any of following conditions are true then an event shall be generated: – line voltage is not active	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

	– the Master Controller is not in neutral position (coast) – train speed is not standstill – traction is not possible (blocked) – DEP brake is not selected				
2F_09.03.Zefiro-Europe.CONCEPT.465	The washing mode shall be aborted if any of following conditions are true: – emergency brake active – available traction is less than 25%. – virtual push button is pressed again – brake mode is set to "INDIRECT" (IEP/IP) When master controller (or DBV) is moved in coasting/braking washing mode shall remain active (the train shall go in coasting or shall brake in a normal way)	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_09.03.Zefiro-Europe.TRS.190
Description	It shall be possible for the speed at which the train passes through the train washing plant to be adapted to suit each washing plant, i.e. between 2 and 5km/h.
Safety level	N/A

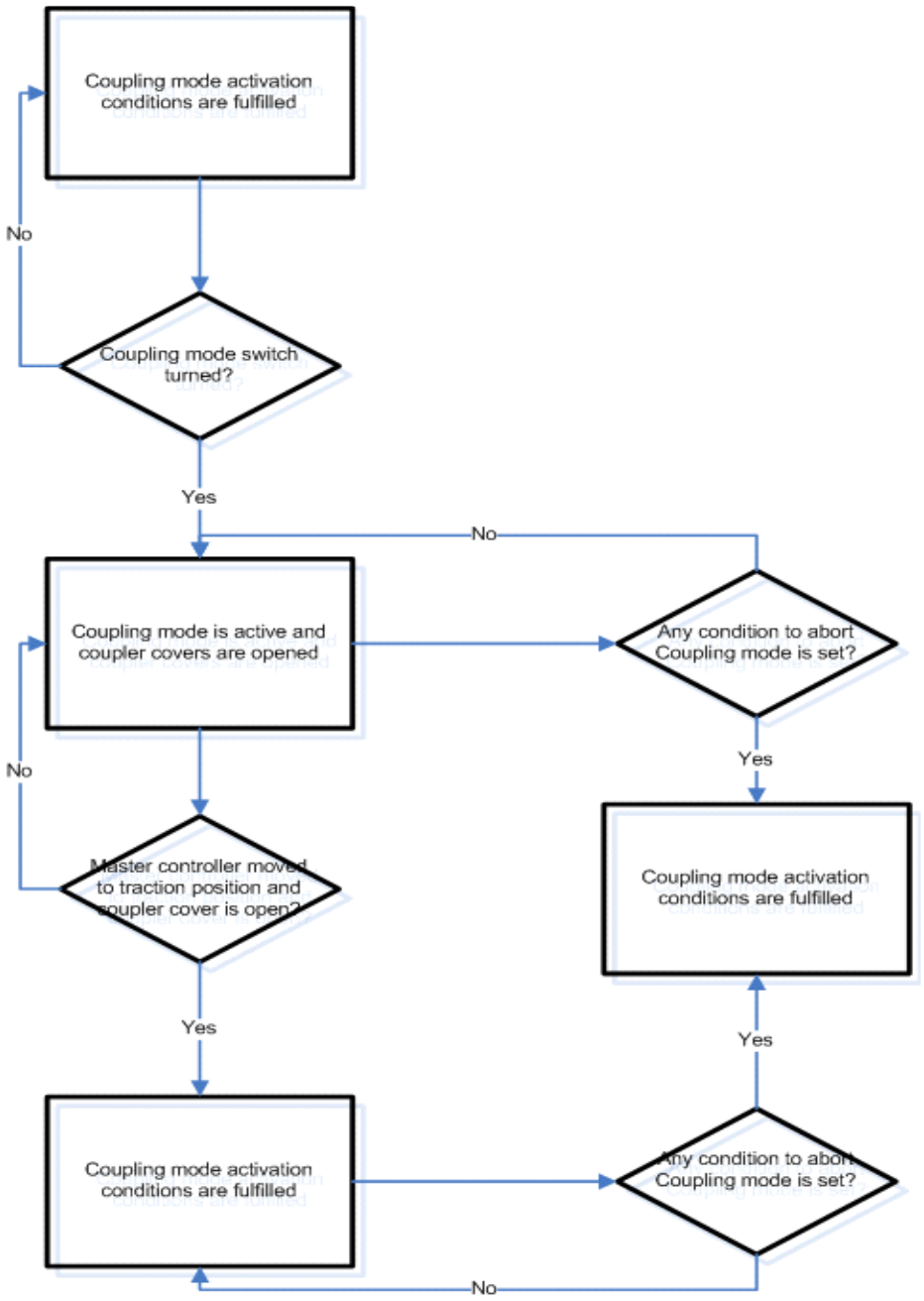
Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1137	If washing mode is active and the master controller is moved to any traction position then traction demand shall be calculated to run the train at speed of 5 km/h.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.459	It shall be possible to adjust the target speed for washing speed using the speed lever. The allowed values shall be between 2 and 5 km/h. The default value shall be 5 km/h.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1103	If the train speed is 5km/h greater than the target speed during washing mode then washing mode shall be aborted.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1138	If the washing speed is aborted due to overspeed then an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1106	If the washing mode is aborted then traction block shall be applied and the traction block shall not be reset until acknowledged with master controller (moved to neutral or braking position).	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

3.1.22 Coupling speed mode

3.1.22.1 *Design*

In coupling mode one 8 car train is driven towards another 8 car train for multiple coupling. The speed is regulated until mechanical coupling is detected. The speed will be 1 km/h to ensure that the couplers does not take damage.

Below flowchart shows how Coupling mode is selected/aborted and the coupling speed control.



3.1.22.1.1 Normal condition

If conditions for Coupling mode are set, then it is possible for the user to activate the Coupling mode by pressing the pushbutton on the driver's desk. If Coupling mode is successfully activated, then the coupler covers are opened and then user moves the master controller to any traction position. The train will then start to move at constant speed of 1km/h.
During the coupling sequence, till when coupling is detected, the coupling lamp shall flash
The movement of the train and Coupling mode will be aborted if coupling pushbutton is pressed again or other condition are not fulfilled.
Activation of coupling speed will not be possible when reverse direction is selected.

3.1.22.1.2 Failure cases

- The speed limitation is not applied even if the Coupling mode has been selected. The speed will be greater than intended when coupling to the other train.
- Coupling mode is not activated by turning the Coupling mode switch. The speed must be set manually by the driver.
- Coupling mode is activated without pressing the dedicated button. Train speed will be limited to 1km/h and coupling covers will be opened.
- Coupler covers are not opened but indicated as opened. The coupler cover will be damaged.

3.1.22.1.3 Failure detection

- Done by the driver/maintainer and Traction block will be applied and an event will be set.
- Done by the driver/maintainer via TOD. The Coupling mode will not be indicated as active on the TOD.
- Done by the driver/maintainer (Speed reduction noted)
- Must be ensured by the driver that coupler covers are open. No detection by the train is possible.

3.1.22.1.4 Failure handling

1. Do not perform the coupling procedure, maintenance needed
2. Do not perform the coupling procedure, maintenance needed
3. If the train is not in service proceed with maintenance, if the train is in service tow could be needed.
4. Coupling to other train not possible. Multiple operation will not be possible and maintenance is required.

3.1.22.2 *Requirements*

ID	2F_09.03.Zefiro-Europe.TRS.219
Description	There shall be a selector switch on the Driver's desk to select coupling speed control.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.144	There shall be a spring loaded push button on the Driver's desk to select coupling speed control.	Derived Requirement	4S_08.07.01.-.Drivers Desk	HW HMI	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1234	When the Coupling mode is active it shall be indicated on the TOD.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1558	The coupling lamp shall flash when coupling mode is active When TCMS detects status "train mechanically and electrically coupled" the lamp shall be lit steadly. The lamp shall be switched off after the driver has acknowledged the new train configuration	Derived Requirement	4S_09.03.05.-.TCMS Software	HW HMI	SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1235	If the Coupling speed mode pushbutton is pressed then TCMS shall be notified.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

ID	2F_09.03.Zefiro-Europe.TRS.86
Description	Coupling speed control shall only be possible to be select when: 1. the Master Controller is in position coast AND 2. train speed is 0 AND 3. traction is possible
Safety level	N/A

Front flaps need to be opened manually in the standing train (without active cab).

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.295	Coupling mode shall be activated if: 1. coupling mode pushbtton pressed AND 2. cab is active AND	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

	3. the Master Controller is in neutral position (coast) AND 4. train speed is 0 (standstill) AND 5. traction is possible (not blocked) AND 6. line voltage is active AND 7. Forward direction is selected AND 8. DEP brake available and selected (if not coupling speed mode is not allowed) AND 9. Parking Brake not applied				
2F_09.03.Zefiro-Europe.CONCEPT.1220	If coupling mode pushbutton is pressed and any of following conditions are true then an event shall be generated for the driver: 1. the Master Controller is not in neutral position (coast) 2. train speed is > 0 km/h 3. traction is blocked 4. line voltage is not active 5. forward direction is not selected 6. DEP brake is not selected 7. Parking Brake applied	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.474	Coupling mode shall be aborted if any of following conditions are true: 1. emergency brake active 2. coupling is detected (mechanical OR electrical) 3. push button is pressed again 4. brake mode is set to "INDIRECT" (IEP/IP) When master controller (or DBV) is moved in coasting/braking coupling mode shall remain active (the train shall go in coasting or shall brake in a normal way)	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1679	DELETED	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1683	**GIVEN** (Vehicle is in "*Coupling Mode*") **WHEN** (Both covers of the DM enabled car aren't "*open*" and "*locked*") **THEN** (Traction Block shall be invoked by TCMS)	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.670	When coupling mode is active the coupler cover in the front shall be opened.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_09.03.Zefiro-Europe.TRS.323
Description	Minimum speed shall be 0.5 km/h and maximum speed shall be 3km/h.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.641	If the train speed exceeds 3 km/h during coupling mode then coupling mode shall be aborted and an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1411	If the coupling mode is aborted then traction block shall be applied and the traction block shall not be reset until until acknowledged with master controller (moved to neutral or braking position).	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1193	If coupling mode is active and the couple cover in the front is open and the master controller is moved to any traction position then traction demand shall be calculated to run the train at speed of 1 km/h.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.302	If electrical coupling is detected, ETB inauguration shall start immediately.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

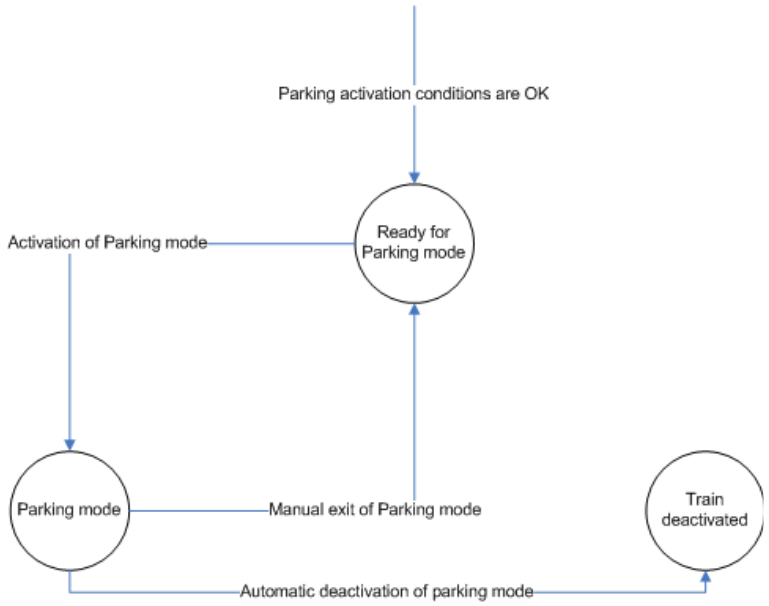
ID	2F_09.03.Zefiro-Europe.TRS.396
Description	A new train configuration shall be always acknowledged by the driver
Safety level	

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1570	Current train configuration shall be displayed on TOD when the train is enabled or everytime the train configuration has changed (e.g copuling/decoupling process, unwanted train separation)	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1569	Driver shall acknowledge the train configuration on TOD, otherwise traction block shall be applied by TCMS	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.1.23 Parking Mode

3.1.23.1 *Design*

Below state diagram defines the fundamentals of the Parking mode



3.1.23.1.1 Normal condition

The Parking mode is activated from the active cab by pressing the Parking push button on driver's desk and at the same time, reset pantographs command and then rotate the driver's key to deactivate the cab. The Parking mode will be confirmed by the illumination of the Parking indicator on driver's desk and an additional indicator visible from outside. All the indicators (internal and external) always show the same state.
Once Parking mode is activated the train will remain all functions that were active before Parking mode. The only difference is that no cab is active and the driver's key is removed.

The Parking mode will normally be deactivated manually from driver's desk by pressing the Parking push button and at the same time set panto switch according to the wanted configuration and then activate the cab by the driver's key. When Parking mode is manually deactivated the train can be resumed in normal operation.

If any of the pre-conditions for Parking mode activation is no longer present during Parking mode the Parking mode will be deactivated automatically after 10 minutes. The train will in this case be also be deactivated by opening of all battery contactors. There will be a time limit how long Parking mode can be active. When the time limit is exceeded Parking mode and the train will be automatically deactivated.

Deactivation of the parking mode occurs also in case the LCB switch is put in "open" position in one the two cabs. In this case the trainset is immediately deactivated.

The Parking mode shall be possible in single and multiple configuration.

3.1.23.1.2 Failure cases

1. Parking mode is not activated due to faulty Parking pushbutton. The train can not be set in Parking mode which means that line voltage will be deactivated when the driver's key must be removed in order to leave the train.
2. Parking mode can not be manually deactivated due to faulty Parking pushbutton.

3.1.23.1.3 Failure detection

1. Detected by the driver/maintainer by the flashing parking indication lamp.
2. Detected by the driver/maintainer by the steady lit parking indication lamp.

3.1.23.1.4 Failure handling

1. Maintenance required to replace faulty parking pushbutton.
2. The Parking mode can be deactivated by pushing the Emergency stop push button in the cab. This will open all LCB's in the train which in turn will trigger the automatic exit of Parking mode.
   Parking can also be deactivated by activating the cab with drivers key.
   Maintenance is required to replace faulty parking push button at next scheduled maintenance.

3.1.23.2 *Requirements*

3.1.23.3 *Parking mode activation*

ID	2F_09.03.Zefiro-Europe.TRS.124
Description	Parking mode shall be entered by: 1. pressing the 'Parking' push button 2. keeping the Parking push button pressed and reset panto switches 3. keeping the Parking push button pressed and rotate and extract the key from the driver's desk 4. releasing the push button; the rolling stock is in Parking mode and the energy conversion systems must remain active depending on the configuration set on the driver's desk prior to entering into Parking mode.
Safety level	SIL 0

ID	2F_09.03.Zefiro-Europe.TRS.125
Description	The release of the push button before the light on it is lit steady, must interrupt the process for entering into Parking and must configure the trainset based on the new position of the driver's desk levers.
Safety level	N/A

ID	2F_09.03.Zefiro-Europe.TRS.126
Description	If the configuration is the initial one, the trainset must behave as if the procedure for entering the Parking mode has not been started.
Safety level	N/A

ID	2F_09.03.Zefiro-Europe.TRS.143
Description	Parking push button shall begin to flash when pressed and shall remain steadily on when the key is removed from the driver's desk.
Safety level	N/A

ID	2F_09.03.Zefiro-Europe.TRS.140
Description	Conditions for entering the Parking mode: 1. train in standstill * 2. Master controller in "0" position on enabled driver desk 3. Speed controller in "0" position on enabled driver desk

	4. DBV in normal position on enabled driver desk 5. Train direction set on "netutral" on enabled driver desk 6. Parking Brake applied * (note: at least 2/3 of parking applied per unit) 7. main reservoir pressure not below 5.5 bar * 8. no fire alarm detected * 9. fire detection system is efficient * 10. no main circuit breaker trip * 11. presence of line voltage * 12. no external power supply inserted * * Conditons above shall be verified on both units in multiple configuration
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.551	Parking mode shall be allowed if: 1. main reservoir pressure not below 5.5 bar * AND 2. no fire alarm is active * AND 3. fire detection system is efficient * AND 4. no main circuit breaker trip * AND 5. line voltage is active * AND 6. train is standstill * AND 7. Master controller is in coast position on enabled driver desk AND 8. Speed controller in "0" position on enabled driver desk AND 9. DBV in normanl position on enabled driver desk AND 10. Parking Brake applied * (note: at least 2/3 of parking brake applied per unit) 11. no external power supply inserted * 12. Train direction set on "netutral" on enabled driver desk * Conditons above shall be verified on both units in multiple configuration	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1263	If the 'Parking' push button is pressed and any of following conditions is set (in single and/or multiple configuration) an advisory event shall be diplayed on the TOD. 1. main reservoir pressure is below 5.5 bar * 2. fire alarm is active * 3. fire detection system is not efficient * 4. main circuit breaker trip * 5. line voltage is not active * 6. train is not standstill * 7. Master controller is not in coast position on enabled driver desk 8. Speed controller not in "0" position on enabled driver desk 9. Parking Brake not applied * (note: at least 2/3 of parking brake applied per unit) 10. external power supply inserted * 11. train direction is not set on "neutral" on enabled driver desk * Conditons above shall be verified on both units in multiple configuration	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.154	Parking mode shall be considered active if all below steps are completed: 1. 'Parking' push button is pressed for at least 3 seconds -> The 'Parking' indicators (internal and external) shall start to flash ONLY IF all preconditons are satisfied 2. The 'Parking' push button is still pressed and pantos switches are put in "low" position	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

	3. The 'Parking' push button is still pressed and the driver's key is rotated to deactivate the cab -> The 'Parking' indicators (internal and external) shall be steady lit in all cabs. 4. The 'Parking' push button is released.				
2F_09.03.Zefiro-Europe.CONCEPT.481	The release of the push button before the Parking indicators lit steady, shall interrupt the process for entering into Parking (on both units in case of multiple configuration).	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.480	There shall be "parking" push button and lamp in the driver's desk.	Derived Requirement	4S_08.07.01.-.Drivers Desk	HW HMI	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1260	The pressed state of the Parking push button shall be indicated to TCMS.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1261	When ordered by TCMS the Parking indication (internal and external) lamp shall be illuminated.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1501	A set of relays (LCAR) that retain the cab active status during the parking mode shall be provided	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1502	LCAR relays shall be supervised indipendently from the other cab active relays. The series of NC contact and the series of NO contact shall be connected to two different TCMS Inputs	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1503	If a fault/incoherence is detected on LCAR relays an event shall be set and it shall not be possible to activate Parking mode till the event is not solved (by driver or maintenance)	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1506	It shall be possible to TCMS to activate parking mode circuits in all cab of the trainset.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1507	After the parking mode enter procedure has been confirmed the TCMS shall enable the parking circuits in all the cab of the trainset (also in multiple operation)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_09.03.Zefiro-Europe.TRS.144
Description	The trainset must have, at the front and back ends, a signal visible from the outside constituted (eg a luminous strip of red LEDs), positioned in the lower centre zone of the front window.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.513	The trainset must have, at the front and back ends, a signal visible from the outside constituted (eg a luminous strip of red LEDs), positioned in the lower centre zone of the front window.	Derived Requirement	4S_08.07.01.-.Drivers Desk	HW HMI	N/A

ID	2F_09.03.Zefiro-Europe.TRS.145
Description	The outside front and back ends visible signal must remain on with a steady light in the Parking mode.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.508	The external 'Parking' indicator shall be steady illuminated when Parking mode is active.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1264	When ordered by TCMS the external Parking indication shall be illuminated.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

ID	2F_09.03.Zefiro-Europe.TRS.131
Description	In parking mode the line voltage shall remain activated.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1265	If the cab is deactivated the line voltage shall be deactivated (open all LCB's and lower all pantographs)	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1266	If the cab is deactivated and Parking mode is set the line voltage shall remain active: 1. LCB shall remain closed 2. also the second pantograph (of the same type) shall be raised	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_09.03.Zefiro-Europe.TRS.132
Description	In parking mode, heating, cooling, saloon lighting and passenger door control shall remain in operation.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1267	If the Parking mode is set the external doors shall remain released.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1268	If Parking mode is set the current status of the interior lighting shall remain.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1269	If Parking mode is set the current status of HVAC shall remain.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.528	The PCU shall receive a status signal that parking mode is active.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.560	To save power and avoid line voltage oscillations: 1. 50% of LCM/CCM, i.e. all propulsion systems without ACM shall shut down when parking mode is active.	Derived Requirement	4S_05.-.Propulsion		N/A

3.1.23.4 *Manual exit from Parking mode*

ID	2F_09.03.Zefiro-Europe.TRS.135
Description	Parking mode shall be finished by: 1. pressing the 'Parking' push button 2. keeping the Parking push button pressed and insert and rotate the key to the driver's desk 3. checking that the light is off and by confirming the exit of the rolling stock from the Parking mode and releasing the push button
Safety level	N/A

ID	2F_09.03.Zefiro-Europe.TRS.147
Description	When the operator exits from parking mode by pressing the Parking push button, it shall flash.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.164	Parking mode shall be switched off if following sequnce is completed 1. The 'Parking' push button is pressed for more than 3 seconds (and shall be kept pressed till end of procedure) and pantograph switches are put in "low" position → The 'Parking indicators (internal and external) shall start flashing; 2. The cab is activated by the driver's key → Train configuration confirmation will be required to the driver on TOD; 3. Train configuration is acknowledged on TOD 4. Any Pantograph switch is turned on	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

	→ TCMS shall acquire as valid panto configuration only if switches status has been stable for 3s; → Once the panto configuration is acquired related LCBs shall be closed (in case of both panto up of the same unit, only rear LCB shall be selected to close); → When LCBs status are confirmed (only one LCB is closed per unit), Panto not selected shall be ordered to lower and 'Parking' indicators (internal and external) shall be switched off in all cabs; 5. The 'Parking' push button is released only if lamp has been switched off This is valid also in case of multiple configuration. Exit parking mode shall be possible from any cab (also from coupled cabs).				
2F_09.03.Zefiro-Europe.CONCEPT.1582	If Parking mode push button is released before the end of exit sequence (lamp still flashing), TCMS shall configure the train with all LCB open and according actual driver desk configuration	Derived Requirement	4S_09.03.05.-.TCMS Software		

After manual exit of the Parking mode, in single or multiple configuration, the cab is active and also line voltage is active. Normal operation of the train can be resumed.

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1409	The Parking mode shall be aborted if one of the following conditions is present: 1. The cab is activated without pressing the parking mode pushbutton 2. More than one key is detected during the enter/exit procedures 3. Supervision of "parking" relays or "cab active retained" relays detects some faults/incoherences 4. Train speed greated than 0 km/h 5. the cab is activated with selected voltage different from catery voltage The TCMS shall re-configure the train opening all LCB and lowering all pantographs.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1500	If the parking mode is aborted and event shall be set on TOD describing the causes and the not fulfilled conditions	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.1.23.5 *Automatic exit from Parking mode*

ID	2F_09.03.Zefiro-Europe.TRS.133
Description	The duration while the train is in parking mode shall be monitored. When the maximum allowed time is reached, the automatic exit shall be triggered (switch to 'train off mode').
Safety level	N/A

ID	2F_09.03.Zefiro-Europe.TRS.134
Description	The maximum allowed time shall be a parameter within the TCMS.
Safety level	N/A

ID	2F_09.03.Zefiro-Europe.TRS.137
Description	In the absence of at least one of the conditions for entering into Parking mode, the Parking mode must be automatically switched off (on each unit in case of multiple configuration).
Safety level	N/A

ID	2F_09.03.Zefiro-Europe.TRS.309
Description	Conditions for leaving the Parking mode: 1. main reservoir pressure below 5.5 bar * 2. train speed greater that 0 km/h * 3. parking brake released * (note: not less than 2/3 of parking brake applied per unit) 4. fire extinguish system is activated * 5. fire detection system is not efficient * 6. main circuit breaker trip * 7. no line voltage * 8. external power supply inserted * * Related to each unit in case of multiple configuration
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.483	Parking mode shall automatically be switched off if: 1. main reservoir pressure below 5.5 bar * 2. main circuit breaker trip (except in case of emergency exit) * 3. fire extinguish system is active (fire detected) * 4. fire extinguish system is not efficient * 5. parking brake released * (note: not less than 2/3 of parking brake applied per unit) 6. no presence of line voltage * 7. more than 1 week have elapsed since Parking mode activation * Related to each unit in case of multiple configuration In case condition 3 or 4 is verified, LCB shall be opened before starting the exit timer In case the LCB of a unit of the multiple configuration is undue opened, the LCB of the other unit shall be opened.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.339	The maximum allowed time in Parking mode shall be a parameter stored i NVRAM.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_09.03.Zefiro-Europe.TRS.138
Description	Upon automatic exit from Parking mode, the following must occur: 1. the start of the 10 minute timed period; 2. the activation of the internal and external flashing signals;
Safety level	N/A

ID	2F_09.03.Zefiro-Europe.TRS.139
Description	During the timed period, the trainset must remain with the panto up and, after 10 minutes, the following must occur, in succession: 1. controlled closure (therefore with sensor edge switched on) of all of the access doors, 60 seconds prior to the expiration of the timed period; 2. cut-off of the train batteries 3. train enters train OFF mode
Safety level	N/A

ID	2F_09.03.Zefiro-Europe.TRS.146
Description	The external Parking signal must flash upon automatic exit from Parking mode (for example when the protections are triggered) and during the timing prior to the cut-off of the train batteries.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.485	If parking mode is pending to be automatically switched off depending on the condtions not beeing fulfilled, this shall be indicated by: 1. the 'Parking' indicators shall flash.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1272	The automatic shut-down of the train shall be possible to stop if the Parking mode is manually turned off.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.489	When 9 minutes has elapsed since trigger to automatically exit Parking mode was detected the release of the external doors shall be removed.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1273	When 10 minutes has elapsed since trigger to automatically exit Parking mode was detected the following shall happen: 1. all LCB's in the train shall be opened (if not already open) 2. all pantographs in the train shall be lowered 3. all battery contactors in the train shall be opened.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1571	If Automatic exit procedure is triggered by the following condition: 1. No Line voltage detected in case during the following 9 minutes the line voltage is detected again and only if all the other preconditions are still satisfied, the automatic procedure shall be stopped and LCB shall be re-closed. Consequently the train status "parking mode" is reactivated. LCB shall be automatically closed after a defined amount of time from the line voltage return that is proportional to the train number (to avoid simultaneous closing of many train LCBs)	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

3.1.23.6 *Emergency Exit*

ID	2F_09.03.Zefiro-Europe.TRS.384
Description	If the driver commands the opening of the LCB from one of the cabs, the immediate exit from the parking mode shall be commanded. The train will go into the "disabled" state with all LCB open and all pantographs down
Safety level	

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.513	If the driver commands the opening of the LCB from one of the cabs or any Emergency Push button is pressed from one of the cabs the immediate exit from the parking mode shall be commanded (no timing to disable battery will start) The train will go into the "disabled" state with all LCB open and all pantographs down	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.1.24 HMI Power Reset

3.1.24.1 *Design*

If the HMI safe processor detects inconsistencies (screen supervision, watch dog or SDT error of input data) it enters the operational safe state (DOSS). From this state it can be recovered by a power cycle. It cannot be recovered from a non operational safe state (NOSS). If TCMS detects that HMI is in fail safe state an automatic power cycle shall be performed max two times, then the HMI shall be replaced.

3.1.24.1.1 Normal condition

In normal operation the HMI is working properly and TCMS is not commanding any power cycle. The power supply is connected through a relay normally closed contact.
In case an inconsistency is detected TCMS energizes the relays disconnecting HMI power supply

3.1.24.1.2 Failure cases

1. Command relay is kept energized (TOD not supplied)
2. Command relay faulty (power cycle not possible)

3.1.24.1.3 Failure detection

1. detected by the Driver during operation
2. detected by the Driver during operation

3.1.24.1.4 Failure handling

1. Command relay shall be replaced during maintenance
2. Replace TOD during operation. Command relay to be replaced during maintenance

3.1.24.2 *Requirements*

ID	2F_09.03.Zefiro-Europe.TRS.395
Description	HMI is in fail safe state shall be automatic recovered by TCMS via Power cycles
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1555	A RIO DO shall control a relay with a normally closed contact in the power supply chain of the TOD . The power shall be cut via an active RIO DO.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		

3.1.25 Train Integration

3.1.25.1 *Design*

3.1.25.1.1 Normal Condition

3.1.25.1.2 Failure cases

3.1.25.1.3 Failure detection

3.1.25.1.4 Failure handling

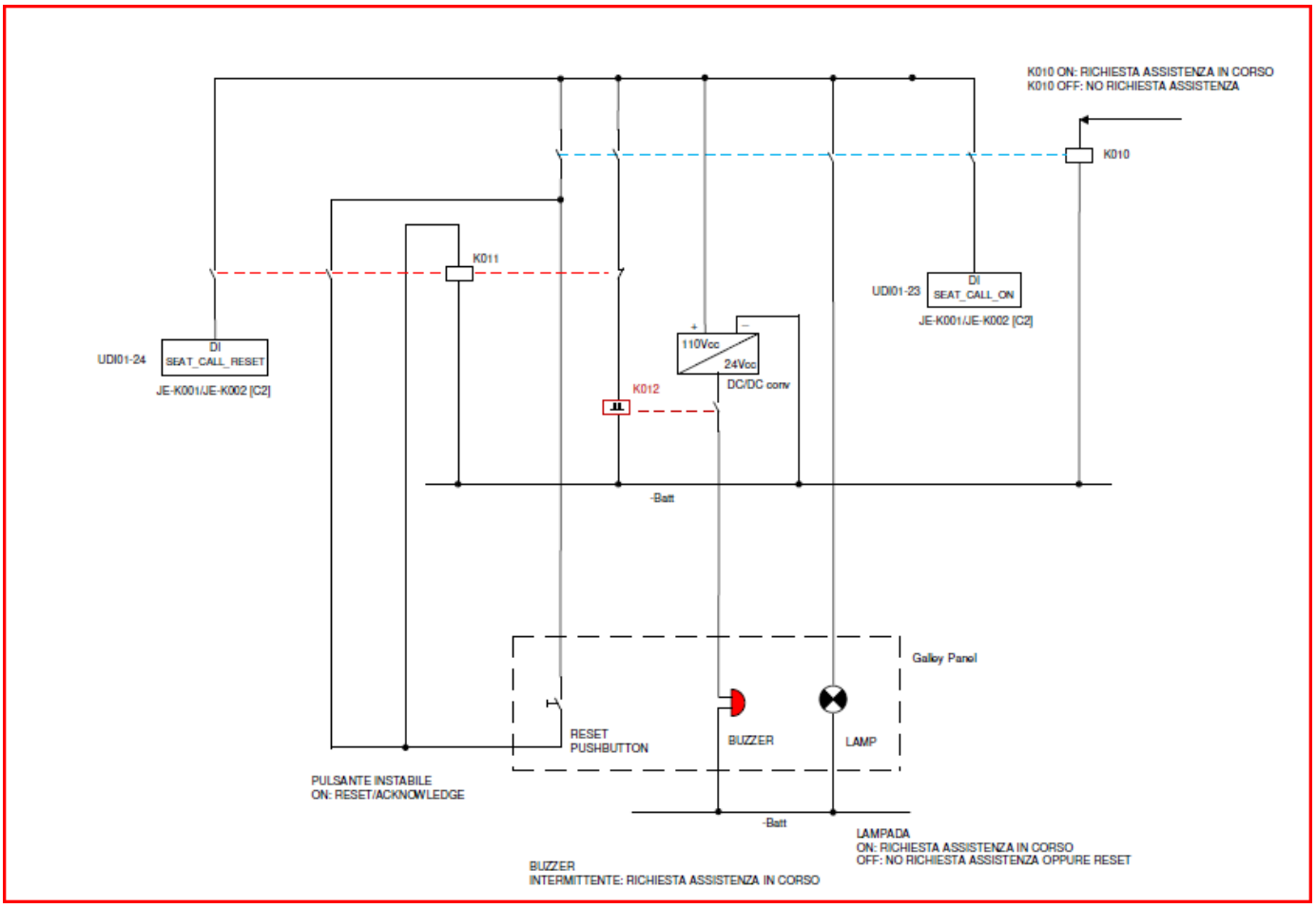
3.1.25.2 *Requirments*

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1608	When Green Loop is de-energised, TCMS shall command Platform Mode to the all Saloon HVACs	Derived Requirement	4S_09.03.05.-.TCMS Software		

3.1.26 Executive seat call for aid

3.1.26.1 *Design*

On each of 14 seat of executive class (DM1 car), a push button is provided to require assistance from on board personnel. The activation of push button on one or more seat cause a local indication on the seats and a global indication (lamp and buzzer) on a panel located in the galley area of DM1 car.
Intervention of on board personnel is necessary to reset both global indication (on galley panel) and local indication (on executive seats).



3.1.26.1.1 Normal Condition

TCMS detect call for aid from one or more seat.
TCMS detect reset from on board personnel of call for aid request

3.1.26.1.2 Failure cases

3.1.26.1.3 Failure detection

3.1.26.1.4 Failure handling

3.1.26.2 *Requirements*

Nr	Description	Applicability Details	Derived to	User Interface	SIL0

2F_09.03.Zefiro-Europe.CONCEPT.HR60	Each executive seat shall be provided of a local lamp to indicate that an assistance call is active and a button to activate and de -activate the call	Derived Requirement	4S_08.04.-.Catering		
2F_09.03.Zefiro-Europe.CONCEPT.HR61	When a push button for assistance call is pressed the local lamp shall be lighted	Derived Requirement	4S_08.04.-.Catering		
2F_09.03.Zefiro-Europe.CONCEPT.HR62	When the assistance call is active and the push button for assistance call is pressed the local lamp shall be switched off	Derived Requirement	4S_08.04.-.Catering		
2F_09.03.Zefiro-Europe.CONCEPT.HR66	When one or more push button of assistance call are pressed: - Indication on galley panel (lamp + flashing buzzer) will be activated - TCMS DI will be activated (SEAT_CALL_ON)	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		
2F_09.03.Zefiro-Europe.CONCEPT.HR63	When one or more push button of assistance call are pressed: - Diagnostic event shall be set indicating the assistance call activation	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_09.03.Zefiro-Europe.CONCEPT.HR54	When reset push button on galley panel will be pressed: - Buzzer on galley panel will be de-activated; - TCMS DI will be activated (SEAT_CALL_RESET)	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		74
2F_09.03.Zefiro-Europe.CONCEPT.HR64	When reset push button on galley panel will be pressed: Diagnostic event shall be set indicating the assistance call reset from personnel	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_09.03.Zefiro-Europe.CONCEPT.HR55	When all the active assistance calls are reset from seats: - Lamp on galley panel will be de-activated; - TCMS DI (SEAT_CALL_ON) will be de-activated	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		
2F_09.03.Zefiro-Europe.CONCEPT.HR65	When TCMS DI (SEAT_CALL_ON) will be de-activated Diagnostic event shall be set indicating that all the assistance calls have been satisfied	Derived Requirement	4S_09.03.05.-.TCMS Software		

4 Signal Interface

4.1 I/O Interface

4.1.1 Supplier system

Nr	Functional name	Type	Linked to requirement
2F_09.03.Zefiro-Europe.CONCEPT.28			

4.1.2 TCMS SW

4.1.2.1 *Input*

Cab Activation/Deactivation:

Nr	Functional name	Type	Linked to requirement
2F_09.03.Zefiro-Europe.CONCEPT.1297	1. 'Any cab active' train line 2. 'Any cab active' train line **(redundant)**	L	2F_09.03.Zefiro-Europe.CONCEPT.948 2F_09.03.Zefiro-Europe.CONCEPT.1253
2F_09.03.Zefiro-Europe.CONCEPT.1299	1. 'Local cab activated' 2. 'Local cab activated' **(redundant)**	L	2F_09.03.Zefiro-Europe.CONCEPT.1006
2F_09.03.Zefiro-Europe.CONCEPT.1298	1. 'Drivers key in active position' 2. 'Drivers key in active position' **(redundant)**	L	2F_09.03.Zefiro-Europe.CONCEPT.1006
2F_09.03.Zefiro-Europe.CONCEPT.1300	1. 'Drivers key in off position' 2. 'Drivers key in off position' **(redundant)**	L	2F_09.03.Zefiro-Europe.CONCEPT.1006

Demand of Driving Direction:

Nr	Functional name	Type	Linked to requirement
2F_09.03.Zefiro-Europe.CONCEPT.1302	1. 'Forward selection pushbutton' 2. 'Forward selection pushbutton' **(redundant)**	L	2F_09.03.Zefiro-Europe.CONCEPT.969
2F_09.03.Zefiro-Europe.CONCEPT.1303	1. 'Neutral selection pushbutton' 2. 'Neutral selection pushbutton' **(redundant)**	L	2F_09.03.Zefiro-Europe.CONCEPT.969
2F_09.03.Zefiro-Europe.CONCEPT.1304	1. 'Reverse selection pushbutton' 2. 'Reverse selection pushbutton' **(redundant)**	L	2F_09.03.Zefiro-Europe.CONCEPT.969

Traction Brake Reference:

Nr	Functional name	Type	Linked to requirement
2F_09.03.Zefiro-Europe.CONCEPT.1308	1. Encoder 1 4-20mA signal	L	2F_09.03.Zefiro-Europe.CONCEPT.1148 2F_09.03.Zefiro-Europe.CONCEPT.1176
2F_09.03.Zefiro-Europe.CONCEPT.1309	1. Encoder 2 4-20mA signal	L	2F_09.03.Zefiro-Europe.CONCEPT.1148 2F_09.03.Zefiro-Europe.CONCEPT.1176
2F_09.03.Zefiro-Europe.CONCEPT.1310	1. 'Master control EB application	L	2F_09.03.Zefiro-Europe.CONCEPT.1028
2F_09.03.Zefiro-Europe.CONCEPT.1316	1. 'Master control back up mode switch	L	2F_09.03.Zefiro-Europe.CONCEPT.1184
2F_09.03.Zefiro-Europe.CONCEPT.1311	1. Speed lever not in "+" position 2. Speed lever not in "+" position **(redundant)**	L	2F_09.03.Zefiro-Europe.CONCEPT.1167 2F_09.03.Zefiro-Europe.CONCEPT.1395

2F_09.03.Zefiro-Europe.CONCEPT.1312	1. Speed lever not in "-" position 2. Speed lever not in "-" position **(redundant)**	L	2F_09.03.Zefiro-Europe.CONCEPT.1167 2F_09.03.Zefiro-Europe.CONCEPT.1395
2F_09.03.Zefiro-Europe.CONCEPT.1314	1. Speed lever in "0" position 2. Speed lever in "0" position **(redundant)**	L	2F_09.03.Zefiro-Europe.CONCEPT.1167 2F_09.03.Zefiro-Europe.CONCEPT.1395
2F_09.03.Zefiro-Europe.CONCEPT.1315	1. Speed lever acknowledgment pushbutton 2. Speed lever acknowledgment pushbutton **(redundant)**	L	2F_09.03.Zefiro-Europe.CONCEPT.1167 2F_09.03.Zefiro-Europe.CONCEPT.1395

Traction Safe:

Nr	Functional name	Type	Linked to requirement
2F_09.03.Zefiro-Europe.CONCEPT.1404	Brake pipe pressure switch 1		2F_09.03.Zefiro-Europe.CONCEPT.1396
2F_09.03.Zefiro-Europe.CONCEPT.1405	Brake pipe pressure switch 2		2F_09.03.Zefiro-Europe.CONCEPT.1396

Traction Safe Override:

Nr	Functional name	Type	Linked to requirement
2F_09.03.Zefiro-Europe.CONCEPT.1320	1. Door interlocking bypass	L	2F_09.03.Zefiro-Europe.CONCEPT.1117
2F_09.03.Zefiro-Europe.CONCEPT.1321	1. Traction safe override	L	2F_09.03.Zefiro-Europe.CONCEPT.1117

Traction Demand in Emergency Mode:

Nr	Functional name	Type	Linked to requirement
2F_09.03.Zefiro-Europe.CONCEPT.1323	1. 'Emergency operation mode' switch	L	2F_09.03.Zefiro-Europe.CONCEPT.1120 2F_09.03.Zefiro-Europe.CONCEPT.1121
2F_09.03.Zefiro-Europe.CONCEPT.1324	1. 'Emergency operation mode' trainline	L	2F_09.03.Zefiro-Europe.CONCEPT.1120 2F_09.03.Zefiro-Europe.CONCEPT.1121
2F_09.03.Zefiro-Europe.CONCEPT.1325	1. 'Traction' Emergency operation mode trainline	L	2F_09.03.Zefiro-Europe.CONCEPT.1049 2F_09.03.Zefiro-Europe.CONCEPT.1124
2F_09.03.Zefiro-Europe.CONCEPT.1326	1. 'Brake' Emergency operation mode trainline	L	2F_09.03.Zefiro-Europe.CONCEPT.1049 2F_09.03.Zefiro-Europe.CONCEPT.1124

Driver Desk supervision:

Nr	Functional name	Type	Linked to requirement
2F_09.03.Zefiro-Europe.CONCEPT.1329	Lamp Test pushbutton	L	2F_09.03.Zefiro-Europe.CONCEPT.104

Towing activation:

Nr	Functional name	Type	Linked to requirement
2F_09.03.Zefiro-Europe.CONCEPT.1331	Towing not active selected	L	2F_09.03.Zefiro-Europe.CONCEPT.1236 2F_09.03.Zefiro-Europe.CONCEPT.1237
2F_09.03.Zefiro-Europe.CONCEPT.1332	Towing active selected	L	2F_09.03.Zefiro-Europe.CONCEPT.1236 2F_09.03.Zefiro-Europe.CONCEPT.1237

Coupling speed mode:

Nr	Functional name	Type	Linked to requirement
2F_09.03.Zefiro-Europe.CONCEPT.1334	Coupling speed switch selected	L	2F_09.03.Zefiro-Europe.CONCEPT.144 2F_09.03.Zefiro-Europe.CONCEPT.1235

Parking mode:

Nr	Functional name	Type	Linked to requirement
2F_09.03.Zefiro-Europe.CONCEPT.1336	Parking push button selected	L	2F_09.03.Zefiro-Europe.CONCEPT.480 2F_09.03.Zefiro-Europe.CONCEPT.1260
2F_09.03.Zefiro-Europe.CONCEPT.1346	Parking lamp turned on/off	L	2F_09.03.Zefiro-Europe.CONCEPT.480 2F_09.03.Zefiro-Europe.CONCEPT.1261

Call for aid (executive DM1 seats):

Nr	Functional name	Type	Linked to requirement
2F_09.03.Zefiro-Europe.CONCEPT.HR56	Seats push button activated / de-activated	L	2F_09.03.Zefiro-Europe.CONCEPT.HR66 2F_09.03.Zefiro-Europe.CONCEPT.HR55
2F_09.03.Zefiro-Europe.CONCEPT.HR57	Galley panel push button activated	L	2F_09.03.Zefiro-Europe.CONCEPT.HR54

4.1.2.2 *Output:*

Serious Fault Indication:

Nr	Functional name	Type	Linked to requirement
2F_09.03.Zefiro-Europe.CONCEPT.31	1. Serious Fault LED 1 2. Serious Fault LED 2 (redundant)	L	2F_09.03.Zefiro-Europe.CONCEPT.1079 2F_09.03.Zefiro-Europe.CONCEPT.1085 2F_09.03.Zefiro-Europe.CONCEPT.1142

4.2 Bus Interface

Propulsion to TCMS SW:
Demanding of driving direction:

Nr	Functional name	Type	Linked to requirement
2F_09.03.Zefiro-Europe.CONCEPT.32	Forward trainline input 1		2F_09.03.Zefiro-Europe.CONCEPT.1280
2F_09.03.Zefiro-Europe.CONCEPT.1349	Forward trainline input 2		2F_09.03.Zefiro-Europe.CONCEPT.1280
2F_09.03.Zefiro-Europe.CONCEPT.1350	Reverse trainline input 1		2F_09.03.Zefiro-Europe.CONCEPT.1280
2F_09.03.Zefiro-Europe.CONCEPT.1351	Reverse trainline input 2		2F_09.03.Zefiro-Europe.CONCEPT.1280

Traction Safe:

Nr	Functional name	Type	Linked to requirement
2F_09.03.Zefiro-Europe.CONCEPT.1352	Traction safe 1 MCM1 is energised		2F_09.03.Zefiro-Europe.CONCEPT.1287
2F_09.03.Zefiro-Europe.CONCEPT.1353	Traction safe 2 MCM1 is energised		2F_09.03.Zefiro-Europe.CONCEPT.1287

2F_09.03.Zefiro-Europe.CONCEPT.1354	Traction safe 1 MCM2 is energised		2F_09.03.Zefiro-Europe.CONCEPT.1287
2F_09.03.Zefiro-Europe.CONCEPT.1355	Traction safe 2 MCM1 is energised		2F_09.03.Zefiro-Europe.CONCEPT.1287

Train Speed Calculation:

Nr	Functional name	Type	Linked to requirement
2F_09.03.Zefiro-Europe.CONCEPT.1360	Axle speed 1 (axle 1, MOCU/M1)		2F_09.03.Zefiro-Europe.CONCEPT.399
2F_09.03.Zefiro-Europe.CONCEPT.1361	Axle speed 2 (axle 1, MOCU/M2)		2F_09.03.Zefiro-Europe.CONCEPT.399
2F_09.03.Zefiro-Europe.CONCEPT.1340	MOCUM_X speed signal	A	2F_09.03.Zefiro-Europe.CONCEPT.973 2F_09.03.Zefiro-Europe.CONCEPT.1130

TCMS SW to Propulsion:
Traction Brake Reference:

Nr	Functional name	Type	Linked to requirement
2F_09.03.Zefiro-Europe.CONCEPT.1356	Traction demand		2F_09.03.Zefiro-Europe.CONCEPT.794
2F_09.03.Zefiro-Europe.CONCEPT.1357	Traction is active		2F_09.03.Zefiro-Europe.CONCEPT.795
2F_09.03.Zefiro-Europe.CONCEPT.1358	Braking is active		2F_09.03.Zefiro-Europe.CONCEPT.796

Traction Safe:

Nr	Functional name	Type	Linked to requirement
2F_09.03.Zefiro-Europe.CONCEPT.1348	Traction safe loop status		2F_09.03.Zefiro-Europe.CONCEPT.1367

Train Speed Calculation:

Nr	Functional name	Type	Linked to requirement
2F_09.03.Zefiro-Europe.CONCEPT.1359	Train speed (for slip/slide control)		2F_09.03.Zefiro-Europe.CONCEPT.860

Speed Limitation:

Nr	Functional name	Type	Linked to requirement
2F_09.03.Zefiro-Europe.CONCEPT.1362	Reference speed for wheel diameter calibration (ATP speed)		2F_09.03.Zefiro-Europe.CONCEPT.1368

Speed Calibration:

Nr	Functional name	Type	Linked to requirement
2F_09.03.Zefiro-Europe.CONCEPT.1363	Calibrate wheel diameter		2F_09.03.Zefiro-Europe.CONCEPT.404

Towing Activation:

Nr	Functional name	Type	Linked to requirement
2F_09.03.Zefiro-Europe.CONCEPT.33	Command: enable towing mode		2F_09.03.Zefiro-Europe.CONCEPT.1238

Braking to TCMS SW:

Nr	Functional name	Type	Linked to requirement
2F_09.03.Zefiro-Europe.CONCEPT.1338	BCU_X speed signal	A	2F_09.03.Zefiro-Europe.CONCEPT.973 2F_09.03.Zefiro-Europe.CONCEPT.1130
2F_09.03.Zefiro-Europe.CONCEPT.1339	WSP_X speed signal	A	2F_09.03.Zefiro-Europe.CONCEPT.973 2F_09.03.Zefiro-Europe.CONCEPT.1130

5 References

[1] FA020022675 Drive Control
[2] FA020023100 Brake Control